Measuring Shortages since 1900*

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Abstract

Shortages—the inability of supply to meet demand at prevailing prices—recur throughout economic history but are hard to measure with traditional data. Using 25 million U.S. newspaper articles, we build a monthly index of shortages of industrial goods, labor, food, and energy since 1900. A structural VAR reveals that a shortage shock produces a persistent rise in inflation, rather than a one-time, transitory price-level shift. It also generates a larger GDP decline than a traditional supply shock for the same rise in inflation—evidence that shortages reflect constraints on price adjustment that limit inflation but magnify output losses.

KEYWORDS: Shortages; Inflation; Textual Analysis; Predictive Regressions; Structural VAR Model.

JEL CLASSIFICATION: C32, C55, E31, N10.

*The data on shortages presented in this paper and monthly updates of the index are available at https://www.matteoiacoviello.com/shortages.html. Caldara and Iacoviello are at the Federal Reserve Board and are both affiliated with the CEPR. Yu is at the University of California Los Angeles. We thank Sam Nate for outstanding research assistance. The views expressed in this paper are solely the responsibility of the authors and should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or of anyone else associated with the Federal Reserve System.

1 Introduction

Macroeconomic models typically assume that buyers obtain what they demand at the prevailing price—either because prices adjust to clear markets or because firms stand ready to meet demand at the prices they have set. Yet over the past century, buyers in many markets have been unable to obtain what they demanded. The most visible cases were the two world wars, the energy crises of the 1970s, and the COVID-19 pandemic. Supply fell short while prices were held back by controls, contracts, or norms against gouging: buyers queued, goods were rationed, and factories idled for lack of inputs. Such shortages are central to historical accounts of these episodes, and rationing was central to the disequilibrium models of the 1970s (Barro and Grossman, 1971), which recent work has revived.[1] Two basic questions nonetheless remain open. How often have shortages occurred over the past century, and how do episodes compare with one another? And are shortages merely a symptom of other shocks, or do they affect inflation and output in their own right?

These questions remain open for a common reason: shortages are hard to measure. Prices and quantities record transactions that took place; a shortage is a transaction that did not. A family unable to buy enough heating coal in 1918, a driver unable to buy gasoline in 1973, or a shopper facing empty shelves in 2021 would appear in economic statistics, if at all, only as lower spending, hard to distinguish from a voluntary cutback. Existing indicators—such as job vacancies and order backlogs—capture pieces of the problem but none provides a long-run measure of demand that goes unmet across the economy. Shortages do, however, surface directly in contemporary newspaper reporting—in reports of gasoline lines, unfilled orders, and scarce workers.

This paper builds a measure of shortages from newspaper reporting and uses it to answer both questions. We find that shortages have been a recurring and often intense feature of the U.S. economy since 1900. Shortages respond to demand and supply shocks, but they also shape macroeconomic outcomes in their own right: higher shortages forecast persistently higher inflation, and, compared with a conventional supply shock, a shortage shock has more persistent inflationary effects and lowers output by more for the same rise in inflation.

[1] On shortages under wartime price controls, see Galbraith (1952) and Rockoff (1984); on rationing during postwar oil shocks, see Hamilton (2011) and Ramey and Vine (2011); on shortages in the pandemic-era inflation, see Bernanke and Blanchard (2025). Bénassy (1993) surveys the early disequilibrium literature; Barro (2025), Holden (2025), Flynn et al. (2026), and Liu (2026) are recent efforts to bring rationing back into macroeconomic models.

In Section 2, we build a shortage index covering 1900 through 2025 from about 25 million articles in six major U.S. newspapers. An article counts toward the index if it mentions a shortage term—such as 'shortage,' 'scarcity,' 'bottleneck,' or 'rationing'—in close proximity to a term from one of four categories: energy, food, industrial goods, or labor. These are the markets in which newspapers most often report shortages, and together they cover most of what the economy uses in production and consumption. Each category yields its own sub-index, and the four sum to a headline index. Thus, episodes in which several markets experience shortages at once register as the most intense.

The index shows that shortages are episodic: long quiet stretches are punctuated by bursts during the two world wars, the 1970s, and the pandemic, when it reached several times its historical average. Between these bursts, the index records smaller episodes with very different causes—coal shortages during strikes in the early 1900s, drought in the 1930s, and labor shortages in the late 1960s. The four decades from 1980 to 2019 were unusually free of shortages, so the pandemic marked the return of an old phenomenon rather than the arrival of a new one. Episodes also differ in which markets they affect: the world wars produced shortages across all four categories at once, the 1970s were dominated by energy, and the pandemic by labor and industrial goods. Country sub-indexes show a mix of global and local episodes, such as the oil shortages in France and the United Kingdom after the 1956 Suez Crisis and in Japan after the 2011 tsunami.

The index measures how intensely shortages are discussed in the news, not shortage of quantities directly. Thus, it is potentially an imperfect proxy: news coverage need not track actual shortages, and the word "shortage" is sometimes used figuratively or in senses outside our scope. In Section 3, we validate that the index is nonetheless a good proxy for the incidence of shortages in two ways. First, through a precision-recall audit: we draw a random sample of about 10,000 articles from the full corpus and classify each independently with a large language model. The index reliably separates genuine economic shortages from superficially similar mentions, and the benchmark query favors precision over recall—loosening the search would admit more false articles. Second, we compare the index to measures of supply-chain pressure built from freight data, survey responses, or news text. Although these measures capture frictions in delivering goods rather than shortages directly, where the series overlap our index co-moves with each of them during major episodes of supply disruptions. It diverges, however, from the highly cyclical supplier-deliveries index in recessions. Our index also covers

a broader set of markets and extends back to 1900, capturing episodes, such as the two world wars, that no existing measure reaches.

We turn to the macroeconomic effects of shortages in Section 4. We find in predictive regressions that higher shortages forecast persistently higher inflation, not only in the aggregate but across GDP components. This relationship holds over the past century and is robust to augmenting the regressions with commodity prices, wages, and other conventional indicators of inflationary pressures. In addition, each of the four categorical sub-indexes predicts the sector-specific prices and wages most exposed to that category. The relationship with real activity is more variable: significantly negative through the late twentieth century, it is positive during World War II and the COVID-19 pandemic—episodes in which shortages coincided with strong demand pressures against tight supply.

The persistent inflationary effects of shortages, and their shifting relationship with activity, motivate the structural VAR in Section 5, which decomposes the joint movements of inflation, GDP, and shortages into demand, supply, and shortage shocks. Our identification rests on two assumptions. First, demand and supply shocks are identified from the slopes of the economy-wide demand and supply curves and can move prices, activity, and shortages on impact. Second, shortages drive a wedge between desired demand and realized output, a feature typically absent from models of aggregate demand and supply. The wedge varies in response to demand and supply shocks as well as to shortage shocks. Following Baumeister and Hamilton (2019), we use informative but deliberately wide priors rather than exact exclusion restrictions.

Shortage shocks typically arise when two conditions occur together: an event that constrains supply—extreme weather, a natural disaster, a geopolitical conflict, a strike, or a production failure—and a friction that keeps prices from adjusting. The friction most strongly associated with shortages in our data is price controls, though contracts, norms against price increases, and coordination failures can also play a role.[2] A supply shift under flexible prices registers mainly as a supply shock, with prices bearing much of the adjustment. When frictions bind unusually tightly, the model also records a shortage shock.

While traditional demand and supply shocks contribute to movements in the index, shortage shocks are its dominant driver. The effects of shortage shocks on inflation and GDP build

[2] The index's largest spikes tend to fall in periods when price controls or rationing were in force, such as the world wars and the 1970s. As Figure 3 in Section 2 shows, shortages co-occur far more strongly with price controls than with any other phenomenon we examine.

gradually, peak one to two years after the shock, and decay far more slowly than those of a traditional supply shock. For the same rise in inflation, a shortage shock also lowers GDP by roughly three times as much as a supply shock does.

The historical decomposition shows that shortage episodes differ in their causes. In the early 1950s, shortages reflected both demand pressures from the transition to a peacetime economy and shortage-specific disruptions around the Korean War. In the 1970s, shortages rose beyond what the adverse supply shocks of the oil embargo alone can explain, consistent with accounts that stress the role of rationing and allocation rules in the oil shocks of that decade.

According to the model, the rise in shortages and inflation that started in 2020 was driven mainly by supply forces—a combination of traditional supply shocks and shortage shocks—with demand playing a secondary role. This conclusion does not rely on parameters that the pandemic itself may have shaped: when we estimate the model on data through 2019, the decomposition is essentially unchanged. The unwinding of these shocks from 2022 onward then brought inflation down. These results connect to the debate over the drivers of post-pandemic inflation. Giannone and Primiceri (2024) and Bergholt et al. (2026) find that the rise in inflation was mostly driven by demand, while Ascari et al. (2024), Comin et al. (2023), and Shapiro (2026) emphasize a significant role for supply. Our model reproduces the demand-driven account when the supply curve is restricted to be steep, but within our framework the data favor a flatter supply curve and, with it, a supply-driven account.

Our approach builds on previous work combining newspaper sources and textual analysis to measure economic phenomena, such as the Economic Policy Uncertainty Index (Baker, Bloom, and Davis, 2016) and the Geopolitical Risk Index (Caldara and Iacoviello, 2022). Some researchers have developed news-based indicators of shortages (Lamont, 1997; Chen and Houle, 2023) or supply bottlenecks (Burriel et al., 2024) for shorter samples. Others have built measures of supply-chain pressure from transportation and shipping data (Benigno et al., 2022; Bai et al., 2026; Liu et al., 2025; Känzig and Raghavan, 2025). Ours is the first shortage index spanning more than 125 years, with sector- and country-specific sub-indexes, and we use it to study both the causes and the consequences of shortages. Using Google searches for the term "shortage," Bernanke and Blanchard (2025) find that shortages had large but largely transient effects on pandemic-era inflation. Over 125 years of data, we find that these effects are far more persistent.

2 The Shortage Index

In this section, we discuss the construction of the shortage index and show how it evolves around key episodes in U.S. history. We then examine the phenomena with which shortages co-occur and present a breakdown of the index by country.

2.1 Construction of the Index

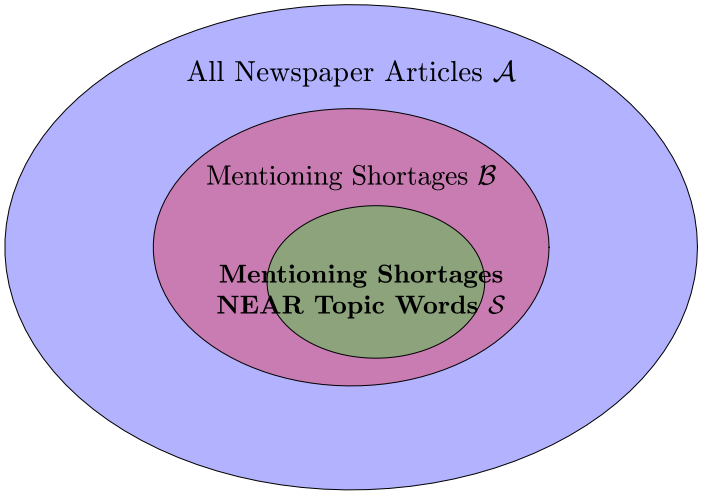
The shortage index is a monthly news-based indicator that measures the intensity with which shortages in industrial goods, labor, food, and energy are discussed in the U.S. press. It is constructed from a sample of roughly 17,000 news articles per month, spanning from 1900 through 2025—about 25 million articles in total. These articles are published in major U.S. newspapers: the *Boston Globe*, the *Chicago Tribune*, the *Los Angeles Times*, *The New York Times*, *The Wall Street Journal*, and *The Washington Post*. Each month, the index counts articles that discuss shortages—the set $\mathcal{S}$ visualized in the top panel of Figure 1—normalized by the total number of articles, denoted as the set $\mathcal{A}$. We isolate these articles with a keyword-based search query, an approach that is transparent, fully replicable, and inexpensive to run across such large dataset. A potential shortcoming of the keyword search approach is that it is context-blind: it cannot tell a reported shortage from a denial ("no shortage of…"), a figurative use, or a shortage outside our scope. This is why we audit it against a large language model in Section 3. Below, we outline the steps used to create the search query, reported in Table 1, for isolating set $\mathcal{S}$.

First, we construct a broader set of articles, $\mathcal{B}$, the pink area in Figure 1. Articles in this set must mention at least one shortage term—namely 'shortage,' 'scarcity,' 'bottleneck,' or 'rationing'—in conjunction with an economics-related term, such as 'economy,' 'market,' or 'commerce.' The shortage terms used are those most frequently linked to economically relevant constraints on production or on the availability of goods to consumers.[3] Including an economics-related term reduces the occurrence of false positives—articles mentioning shortages unrelated to economic phenomena, such as shortages of empathy or goodwill.

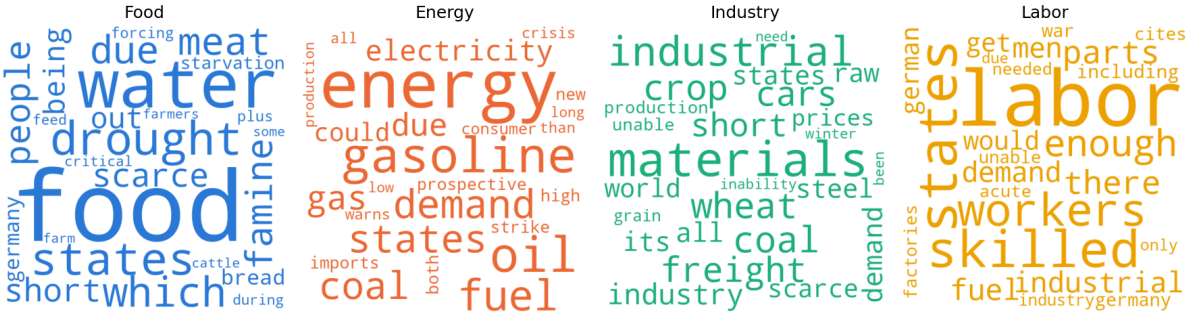
Second, we draw a large random sample of articles from the set $\mathcal{B}$ uniformly across the full

[3] Potential synonyms of shortage, such as 'lack,' 'paucity,' or 'insufficiency,' have a broader range of meanings and are less likely to be specifically associated with economic shortages. In Section 3.1 we formally evaluate a query that adds these synonyms and show that it raises the false-positive rate substantially for only a modest gain in recall, so we exclude them.

Figure 1: Grouping of Newspaper Articles for the Construction of the Shortages Index



(a) Grouping of Newspaper Articles



(b) Word Clouds of the Leading Shortage Topics

Note: Panel (a) shows the sets of articles used to construct the index. Panel (b) shows word clouds for the four leading topics of a Latent Dirichlet Allocation (LDA) fit to the model's short explanations of why each article was classified as a shortage.

1900–2025 period, and extract the 1,000 most frequent collocates within a five-word window of the shortage terms. By construction, these collocates are representative of the entire 125-year period from which the index is built.[4] After removing generic words that carry little information about shortages, we group the remaining collocates—the most common of which include 'oil,' 'coal,' 'water,' 'food,' 'labor,' 'steel,' and 'fuel'—into four topics: energy, food, industry, and

[4] The nature of shortages has of course changed over 125 years: goods such as 'coal,' 'steel,' and 'rubber' appear as shortage collocates in the first half of the sample (1900–1959) but essentially not in the second, while 'energy,' 'staffing,' and 'electricity' become far more prevalent as shortage collocates only in the second half (1960–2025). Selecting the collocates over the entire sample period mitigates any concern about vocabulary drift: the shortage terms themselves are time-invariant, and the four topic categories absorb the changing set of scarce goods.

Table 1: Search Query for the Construction of the Shortage Index

Search Name	**Search Query**	**Peak Month**
Energy Shortages	(*shortages* N/5 *energy*) AND *economics*	Dec-1973
Food Shortages	(*shortages* N/5 *food*) AND *economics*	Mar-1943
Industry Shortages	(*shortages* N/5 *industry*) AND *economics*	May-1942
Labor Shortages	(*shortages* N/5 *labor*) AND *economics*	Jan-2022

Topic Sets	**Components**
shortages	`bottleneck* OR rationing* OR scarcit* OR shortage*`
energy	`anthracite OR blackouts OR coal OR coke OR crude OR diesel OR electricity OR energy OR fuel OR fuels OR gas OR gasoline OR heating OR mines OR oil OR petrol OR petroleum OR pipeline OR refineries OR refinery`
food	`agriculture OR bean* OR beef OR bread OR butter OR cabbage OR cattle OR chicken OR coffee OR corn OR crop OR crops OR drought OR egg* OR famine OR farm OR farmer OR farmers OR fats OR feed OR fertilizer OR fish OR flour OR food OR foods OR foodstuffs OR grain OR lard OR livestock OR meat OR meats OR milk OR pork OR rice OR sugar OR tobacco OR water OR wheat`
industry	`aircraft OR airline OR alloy* OR aluminum OR auto OR automobile* OR automotive OR autos OR car OR cars OR cement OR chip* OR cloth OR clothing OR commodities OR components OR computer OR copper OR cotton OR deliveries OR distribution OR equipment OR freight OR goods OR infrastructure OR iron OR lumber OR machinery OR material OR materials OR metal* OR motor OR paper OR parts OR plane OR planes OR product OR products OR rail OR railway OR rubber OR scrap OR semiconductor* OR shipment* OR shipping OR ships OR silver OR steel OR 'supply chain' OR textile OR textiles OR tin OR tire OR tires OR tonnage OR transit OR transportation OR wood`
labor	`employees OR labor OR labour OR 'man power' OR manpower OR personnel OR professional* OR skilled OR skills OR staff OR staffing OR strike* OR technician* OR training OR union* OR worker OR workers OR workforce`
economics	`budget OR business OR capacity OR commerce OR company OR corporation OR demand OR economic OR economy OR factory OR fiscal OR industr* OR manufactur* OR market OR price OR production OR sales OR suppl* OR tax OR trade OR wages`

Note: The truncation character (*) denotes a search encompassing all possible endings of a word, e.g. "scarcit*" includes "scarcity" and "scarcities." "AND" and "OR" are logical operators, N/5 denotes a proximity operator requiring words appearing within five words from each other. The denominator of the index, the set of all articles $\mathcal{A}$, is isolated with a query requiring common words (the AND be AND to AND of AND and AND at AND in).

labor. The four-topic grouping also finds suggestive support in a Latent Dirichlet Allocation of the free-text shortage explanations returned by the large language model used in our validation exercise in Section 3.1, see bottom panel of Figure 1.[5] We considered three further categories (housing, health, and education) but exclude them: these reflect long-standing social issues more than the frictions we want to capture. Apart from post-WWII housing shortages and the

[5] LDA is a machine learning technique that recovers latent topics from the words that co-occur across documents (Blei et al., 2003). Because the LDA is fit on the model's own explanations, not on article text, it corroborates the taxonomy without independently testing it. We detail the methodology in Appendix B.

COVID-19 health spike, they add little to the headline index (Appendix B).[6]

Using the lists of shortage terms, economic terms and topic-specific terms, we construct the search query shown in Table 1. An article is included in the shortage index (set $\mathcal{S}$ in Figure 1) if it meets two conditions: (1) A shortage term must appear within five words of a topic-specific term, and (2) the article must contain at least one economics term. If an article meets the first condition for several categories, it contributes to each, so the headline index sums the four sectoral sub-indexes. This is by design: an economy short of energy, food, and labor at once is more severely constrained than one short of a single input. Section 3 documents that the requirement of proximity between shortage and topic-specific terms is crucial to reducing false positives.

2.2 Shortages in History

Figure 2a shows the monthly shortage index from 1900 through 2025. The index is calculated as the monthly share of articles discussing economic shortages (number of articles in $\mathcal{S}$ divided by the number of articles in $\mathcal{A}$), scaled so that its average is 100 between 1900 and 2019. Figure 2b presents a breakdown of the index by topic. Table 2 lists the 30 largest spikes in the index.

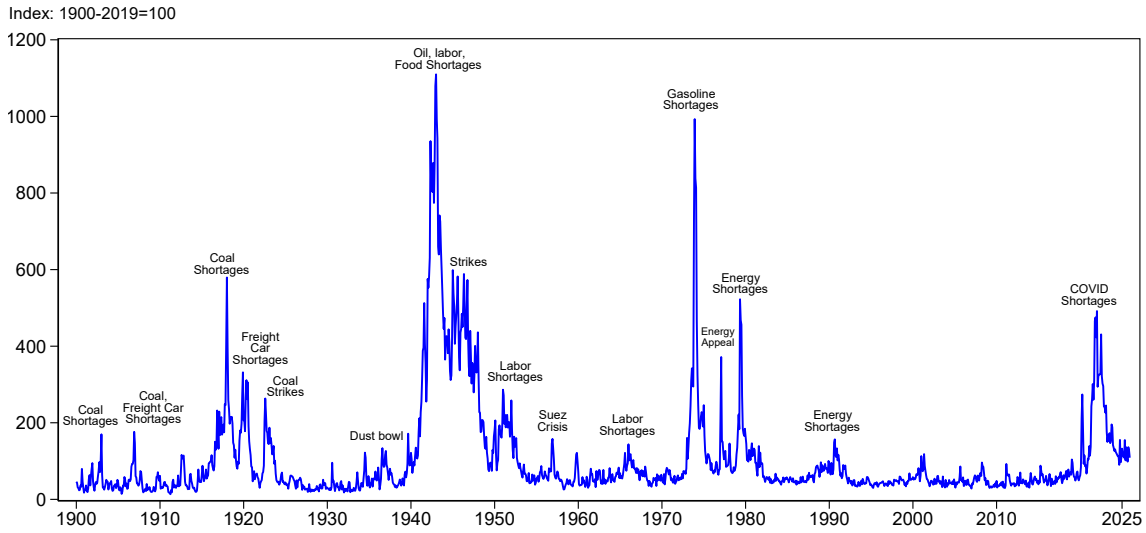
The index varies considerably over time. Adverse geopolitical events, especially wars, are strongly associated with severe shortages, often combining disruptions to supply with frictions, such as price controls, that keep prices from adjusting.[7] The index rises sharply during both world wars, peaking at over 1,000—ten times the sample mean—in January 1943, with spikes recorded across all four categories. Energy shortages are especially tied to the oil shocks of the 1970s, while labor shortages were historically driven by strikes, particularly early in the sample, as in the 1902, 1919, and 1922 coal strikes. The labor sub-index also spikes in each of the four episodes that Michaillat and Saez (2024) identify as periods when the U.S. labor market was inefficiently tight—World War II, the Korean War, the Vietnam War, and the post-pandemic period.

More recently, the index spikes on several occasions during the COVID-19 pandemic. The first spike corresponds to early-pandemic shortages of workers in the health care sector as well

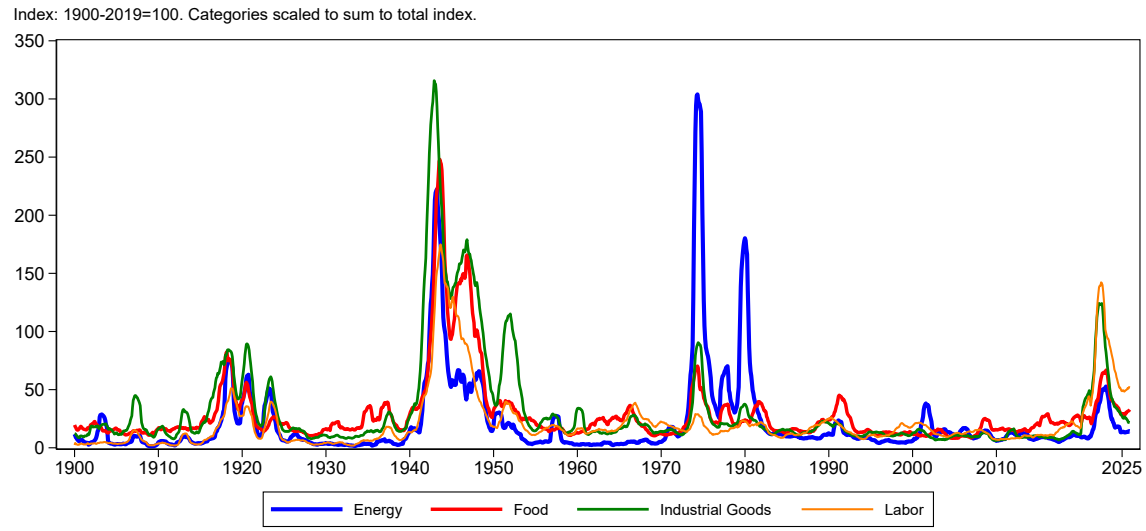
[6] An index that adds housing, health, and education shortages to the four headline categories has a monthly correlation of 0.99 with the headline index over 1900–2025.

[7] Federle et al. (2026) show that wars are associated with supply disruptions both domestically and abroad.

Figure 2: The Shortage Index and Its Categorical Components



(a) The Shortage Index



(b) Decomposition by Category

Note: Panel (a) plots the shortage index from January 1900 through December 2025. Panel (b) shows the four category indexes, scaled to sum to the total index in panel (a).

as shortages of basic household items and groceries caused by initial panic buying. A second, larger spike occurred at the beginning of 2022, driven by global supply bottlenecks as economies reopened after prolonged mobility restrictions in 2020 and 2021. As shown in Figure 2b, this second spike primarily reflected shortages of labor and industrial-goods components.

Table 2: Largest Shortage Spikes, 1900-2025

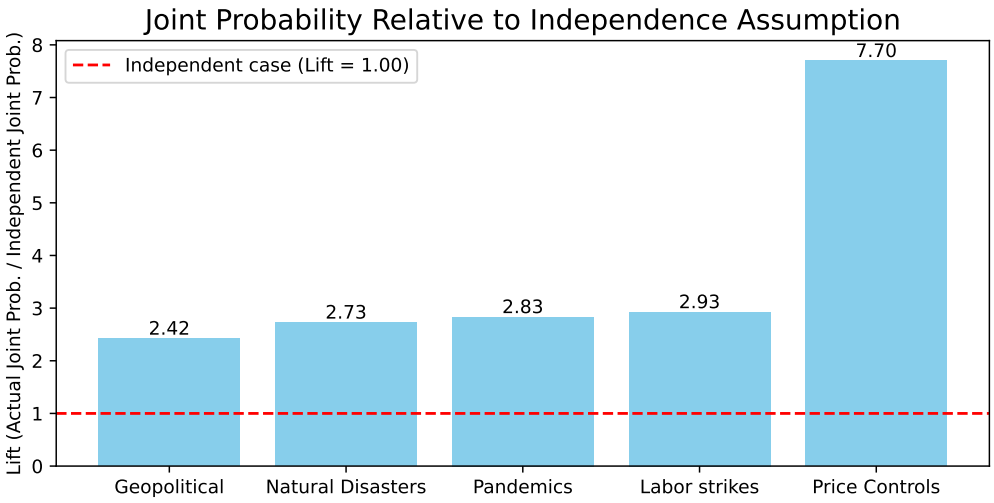
Month	Index	Surprise (st.dev.)	Event
Dec-1901	95	1.14	Nationwide coal shortages
Jan-1903	170	2.83	Nationwide coal shortages
Dec-1906	176	2.37	Shortage of coal and freight cars in Midwest
Nov-1916	232	2.43	Nationwide coal shortages
Jan-1918	579	7.87	Fuel and coal shortages
Dec-1919	332	3.15	Fuel and coal shortages due to war, strikes
Aug-1922	263	3.12	Coal shortage due to strikes
Aug-1930	96	1.39	Drought leads to food and water shortages
Jul-1934	122	1.53	Strike by Teamsters unions in the West Coast
Aug-1936	134	1.53	Dust Bowl shortages
Sep-1939	172	2.22	Steel shortage due to the beginning of World War II
May-1942	936	5.90	War-related energy, materials and labor shortages
Jan-1943	1110	4.96	War-related oil, labor and food shortages
Jan-1945	599	4.52	War-related widespread shortages
Aug-1945	582	3.84	Labor shortages at the end of war
May-1946	589	3.75	Strikes by coal workers and fuel shortages
Jan-1948	436	3.81	Metal, fuel and food shortages
Feb-1950	206	2.06	Coal shortages amid strikes
Jan-1951	286	2.59	Labor shortages due to demand from defense industries
Jan-1952	258	2.70	Nationwide and worldwide shortages
Dec-1956	158	1.65	Oil shortages due to Suez Crisis
Dec-1973	993	10.64	Gasoline shortages due to 1973 oil crisis
Jan-1975	246	1.75	Concerns about gasoline rationing
Feb-1977	372	6.15	Carter's appeal on energy conservation
Feb-1978	146	1.51	Concerns about energy shortages
May-1979	523	4.47	Concerns about energy shortages
Sep-1990	157	1.49	Concerns about energy shortages
Apr-2020	274	4.44	Shortages due to COVID-19 pandemic
Jan-2022	492	2.32	Labor shortages
Oct-2023	196	1.24	United Auto Workers strike

Note: The table lists the 30 largest spikes in the shortage index, 1900–2025. *Index* is the monthly shortage index (1900–2019 = 100). To identify spikes, we regress the index in each month on its values two months earlier and two months later; *Surprise* is the regression residual divided by its standard deviation. We keep only months in which the index is the highest within the 13-month window from $t-6$ to $t+6$, so that each spike is a true local peak, and report the 30 such months with the largest residuals. Event descriptions are our own characterization.

2.3 Co-Occurring Phenomena and Country Indexes

To determine whether shortages tend to occur alongside particular phenomena, we examine their co-occurrence in the news with economic events: labor strikes, natural disasters, adverse geopolitical events, pandemics, and price controls. Figure 3 compares the actual joint probability of these events being mentioned with shortages to the probability expected under independence. The lift ratio, defined as the ratio of the observed joint probability to the product of the

Figure 3: Phenomena Associated with Shortages



Category	*P(Shortage)*	*P(Category)*	*P(Joint)*	*P(Indep.)*
Geopolitical	1.40%	2.66%	0.090%	0.037%
Natural Disasters	1.40%	2.87%	0.109%	0.040%
Pandemics	1.40%	1.27%	0.050%	0.018%
Labor strikes	1.40%	2.70%	0.110%	0.038%
Price Controls	1.40%	0.80%	0.086%	0.011%

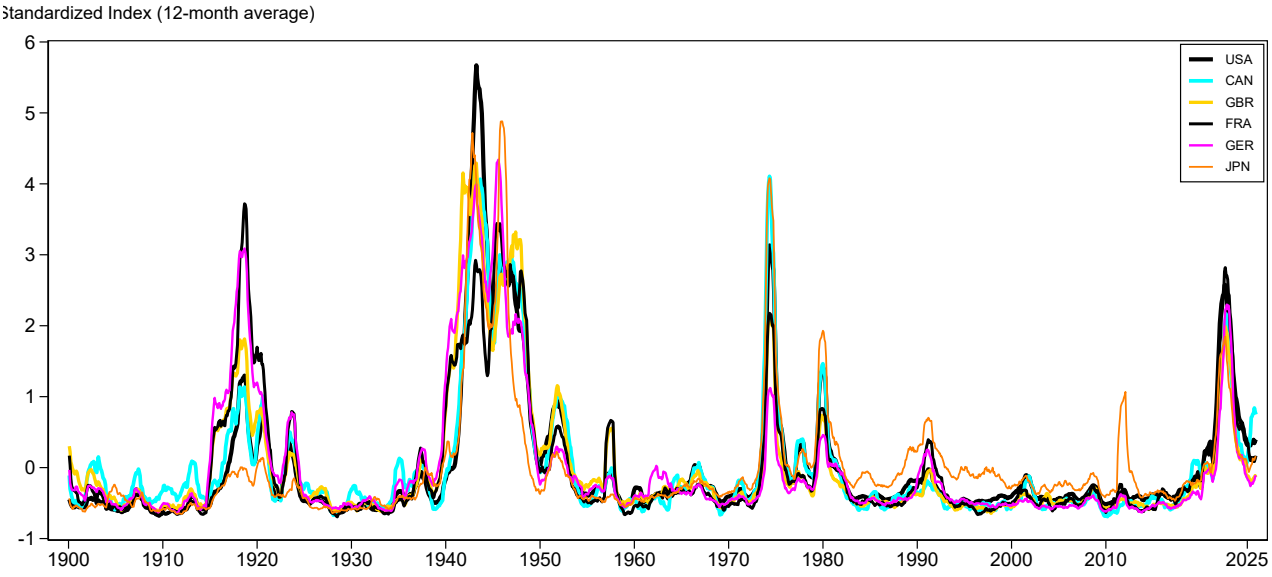
Note: The figure plots the observed probability that an article mentions both shortages and an event category, divided by the probability expected if the two were independent. Higher ratios indicate a stronger association. *P(Shortage)*, *P(Category)*, and *P(Joint)* are the shares of articles mentioning shortages, the category in each row, and both; *P(Indep.)* is the joint probability under independence.

individual probabilities, serves as a measure of association. A lift ratio greater than one indicates that these events co-occur more frequently than would be expected by chance.

Price controls have the strongest association with shortages, co-occurring 7.7 times more frequently than expected under independence. Strikes, natural disasters, geopolitical conflicts, and pandemics show weaker associations, co-occurring two to three times more often than expected. This evidence motivates how we interpret shortage shocks in Section 5: shortages tend to arise when an event that constrains supply coincides with a friction that keeps prices from adjusting, most often price controls.

Finally, Figure 4 slices the index by country, counting articles that satisfy the shortage query and mention a country or one of its major cities. The resulting country indexes spike around shared global events, but differ in timing and intensity: European countries faced earlier, more prolonged shortages than North America during the world wars, and Japan more so than others in the late 1970s. The indexes also recover country-specific episodes—oil shortages in France and the United Kingdom after the Suez Crisis, concerns in Germany

Figure 4: Shortage Index across Countries



Note: Shortage indexes for the United States, Canada, France, Germany, the United Kingdom, and Japan. Indexes are standardized to mean zero and unit standard deviation over 1900 through 2025.

around the 1961 Berlin Wall construction, and shortages following Japan's 2011 tsunami.[8]

3 Assessing the Accuracy of the Shortage Index

We perform two exercises to assess the index's accuracy. First, we audit our keyword-based classification against a large language model, measuring both how many of the articles it flags are genuinely about shortages and how many articles it misses. Second, we compare the index with measures of supply-chain pressure—a related but distinct concept—which also highlights what the index adds relative to existing data.

3.1 Audit Using a Large Language Model

We audit the keyword query against a large language model. We draw a random sample of about 10,000 articles on any topic from the full corpus $\mathcal{A}$, excluding *The Wall Street Journal*, for which full article text was not available. Because the sample covers all articles, not only those the query flags, we can measure both possible errors. The false discovery rate is the

[8] In line with our measurement, the Suez Crisis did not create major energy shortages in Germany because its energy supply was less reliant on Middle Eastern sources compared to France and the United Kingdom.

share of the query's hits that are not shortages; the miss rate is the share of genuine shortage articles the query does not flag. We establish ground truth with Claude Opus 4.8, a large language model that is well-suited for zero-shot and few-shot textual classification in economics. Specifically, we instruct the model, run at high reasoning effort, to read the first 10,000 characters of the article—roughly the first 2,000 words—and to return a single score: 1 if the article mentions an economically relevant shortage, and 0 otherwise.[9]

Ground truth follows our definition of an economically relevant shortage, which excludes housing, medical, education, and military shortages. Such shortages tend to be newsworthy in ways unrelated to the market-clearing frictions we want to capture.[10] We also give the model a fixed set of examples, reproduced in Appendix Table A.1. They include cases that should score zero despite containing the word "shortage," such as denials ("no shortage of…"), figurative uses, and shortages outside our four categories. To check the model's scores, we reviewed 100 articles from the sample, half drawn from those Opus scores 1 and half from those it scores 0. We agree with Opus's score on 98 of the 100.[11]

We evaluate the benchmark query alongside four variants that change one feature: requiring the shortage and topic terms to be adjacent rather than within five words; widening the window to twenty words; dropping the proximity requirement altogether, which yields the broader set $\mathcal{B}$; and, in addition, expanding the list of shortage terms with synonyms such as 'lack' and 'insufficiency.'

Figure 5a plots each query's miss rate against its false discovery rate; lower-left is better. The queries line up along a frontier: every change that lowers one error raises the other. Requiring adjacency lowers the false discovery rate at the cost of more misses, while widening the window, dropping the proximity requirement, or adding synonyms move the other way. Of the 10,296 articles, Claude Opus 4.8 flags 448 as shortages and the benchmark query flags 195,

[9] Truncation economizes on tokens: about 40 percent of articles are longer than the cutoff. We apply the same truncation when computing keyword hits, so the model and the queries see identical text. The prompt defines a shortage as the inability of supply to meet demand at prevailing prices for industry and goods, labor, food and water, or energy, and scores anything else 0. An excerpt: "Return a single binary score: does the article explicitly mention an economic shortage of industry/goods, labor, food/water, or energy, current or prospective? … Score 0 for housing shortages, medical shortages, education-related shortages, financial shortages, and military-related shortages. Score 0 if the shortage is denied ('no shortage of …'). When in doubt, score 0. Use only 0 or 1."

[10] We also drop shortage terms that refer exclusively to military contexts (e.g., "ammunition shortage") from our keyword list.

[11] Cohen's κ is 0.96. In both disagreements Opus scores a shortage that we do not: shoes and equipment for soldiers in 1917, which we treat as a military shortage, and a 2006 book review recounting the Great Leap Forward famine, which we treat as historical rather than current.

of which 162 are also flagged by Opus. Our benchmark query favors precision: about five in six of the articles it flags are genuine shortages, but it misses nearly two-thirds of the shortage articles Opus identifies. Most of the missed articles describe shortages without the terms our search requires.[12]

We accept a high miss rate because the two errors affect the index differently. A roughly constant miss rate simply rescales the index without changing its shape. False positives are more damaging, because they add coverage of unrelated topics that can rise and fall for reasons unrelated to shortages. What matters, then, is whether either error changes over time. Both rates drift over the century—the miss rate falls through the 1970s and rises thereafter, and the false discovery rate rises slowly—but by little compared with the swings in the index itself. In Appendix B we construct an error-adjusted index that corrects for both error rates over time. It is essentially unchanged, with a correlation of 0.99 with the raw index in both levels and annual changes.[13]

A related concern is that, because the index pools several newspapers, it might partly reflect one paper's editorial idiosyncrasies or regional events with little aggregate significance. In Appendix C we plot the shortage index separately for each of the six newspapers. These single-newspaper indexes closely track both the aggregate index and each other.

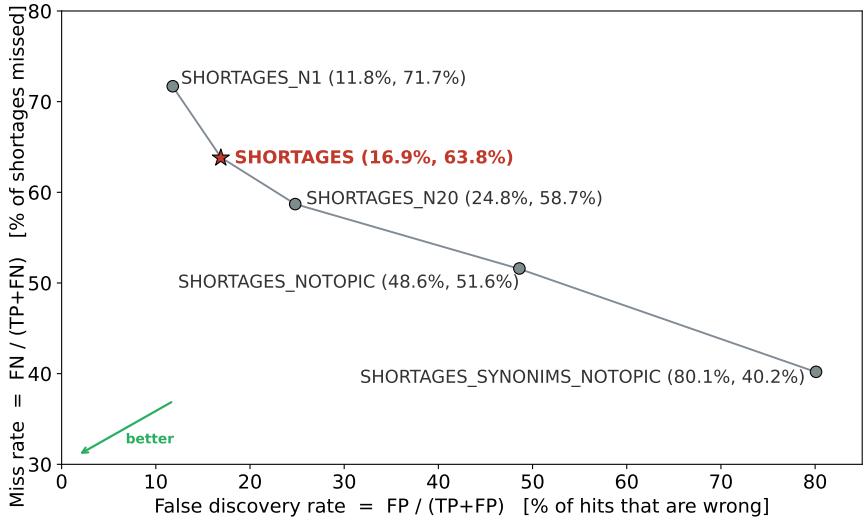
3.2 Comparison with Measures of Supply-Chain Pressure

Figure 5b compares our index with three prominent measures of supply-chain pressure: the New York Fed Global Supply Chain Pressure Index (GSCPI) (Benigno et al., 2022), the ISM Manufacturing Supplier Deliveries Index (SDI), and the Supply Bottlenecks Index (SBI) for the United States (Burriel et al., 2024). The GSCPI combines transportation costs, such as freight rates, with survey measures of delivery times, backlogs, and purchased inventories. The SDI measures supplier delivery delays from purchasing manager surveys. The SBI uses newspaper searches targeting articles mentioning "supply" or "supply chain" alongside disruption terms. These measures capture frictions in delivering goods rather than demand that goes unmet at prevailing prices. The two concepts are related but distinct: shortages often lengthen

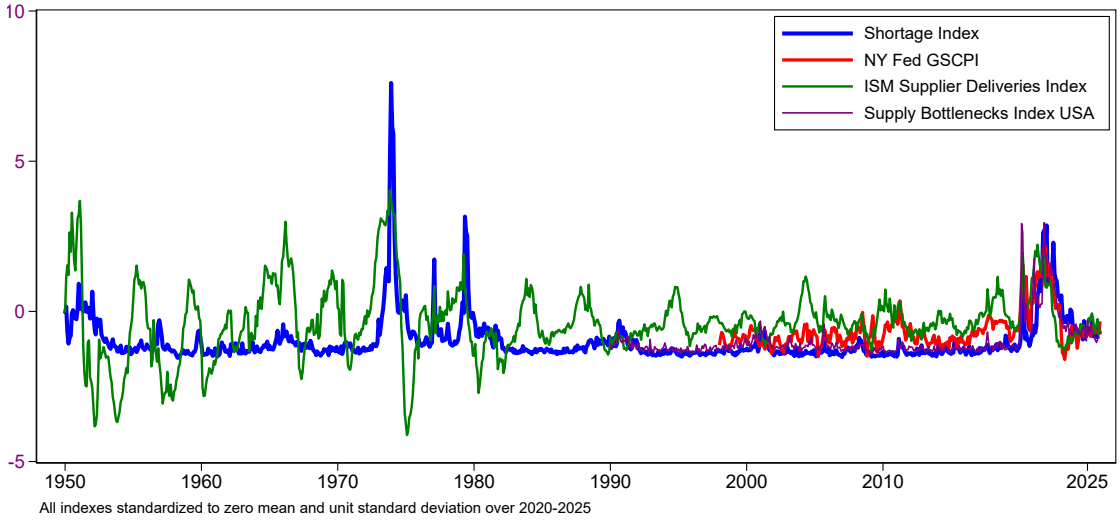
[12] Among the queries we consider, the benchmark attains the highest $F_{0.5}$ score, a standard summary measure of classification accuracy that combines precision and recall but gives precision twice the weight.

[13] Because the error rates come from the labeled sample rather than the full corpus, the error-adjusted index is illustrative, not a precise re-estimate. It runs slightly above the raw index early in the twentieth century, when more shortages were described in language our search does not catch.

Figure 5: Validating the Shortage Index



(a) Accuracy of the Search Query



(b) Comparison to Other Supply Constraints Measures

Note: Panel (a) plots each candidate query's false discovery rate (1−precision) against its miss rate (1−recall), with Opus 4.8 as ground truth. `SHORTAGES` is the benchmark query. Panel (b) compares the shortage index with the GSCPI (Benigno et al., 2022), the ISM Supplier Deliveries Index, and the U.S. Supply Bottlenecks Index (Burriel et al., 2024).

deliveries, but deliveries can also slow while every order is eventually filled, and transport costs rise precisely when markets clear through price. Our index also covers shortages across four sectors—energy, food, industrial goods, and labor—without requiring supply chain framing. We therefore expect it to share some spikes with these measures but not others. We standardize

each variable to have mean zero and unit variance over 2020–2025.

Our index co-moves with these measures during major disruptions. The correlation with the GSCPI is 0.73, with both spiking during COVID-19. Our index correlates at 0.88 with the SBI, over a period dominated by COVID-19—a supply chain disruption affecting multiple sectors simultaneously. This high correlation also reflects the fact that the SBI is built from newspaper text, so both indexes respond to the same news coverage.

The SDI offers a more informative test, since it is built from purchasing-manager surveys. In levels, its correlation with our index is only 0.25, which largely reflects what the SDI measures. Delivery times lengthen whenever demand is strong, so the SDI moves closely with the business cycle whether or not demand goes unmet. The SDI is also a diffusion index. Each month, managers report whether delivery times improved, worsened, or stayed the same, so the index captures the direction of change in supply conditions more than their level.[14] Consistent with this, its correlation with the 12-month percent change in our shortage index rises to 0.48. Beyond these differences in measurement, our sectoral breadth captures episodes not framed in supply chain terms, such as early 20th century labor strikes, wartime price controls and rationing, or geopolitical energy disruptions.

The two series diverge systematically in recessions. Both spiked during the 1973–74 energy crisis, but in the recessions that followed that episode and the 1979 oil shock, the SDI fell well below its average as demand collapsed, while our index declined only gradually toward normal levels. The 1990 Iraq invasion of Kuwait is a further example. Our index and the SBI spike modestly and briefly, while the SDI declines sharply, primarily reflecting the concurrent recession. Energy supply was briefly constrained, but Strategic Petroleum Reserve releases and higher Saudi output prevented prolonged shortages.

A further advantage of our index is historical coverage. The GSCPI begins in 1998, the SBI in 1990, and the SDI in 1948. Our index extends to 1900 and so spans both world wars and the 1970s. This matters because the post-1990 period, apart from the pandemic, saw few severe shortages, limiting the variation available to more recent measures.

[14] If delivery times worsened for half of respondents and improved for the other half, the index would give the same reading as if they were unchanged for all.

4 Shortages as Predictors of Inflation and Activity

In this section, we analyze how shortages relate to inflation and economic activity using a predictive regression framework. Our results indicate that higher levels of shortages are associated with persistently higher inflation, both for the aggregate index and, sector by sector, for each of its four categorical components. Their relationship with economic activity, by contrast, varies over time.

4.1 Predictive Regressions: Shortages, Inflation, and Activity

We explore the relationship among shortages, inflation, and economic activity using the following predictive regression:

$$\Delta Y_{t+h} = \alpha + \beta\, SHORTAGE_t + \sum_{i=0}^{p} \boldsymbol{\gamma}_i' \mathbf{X}_{t-i} + \varepsilon_{t+h} \tag{1}$$

where

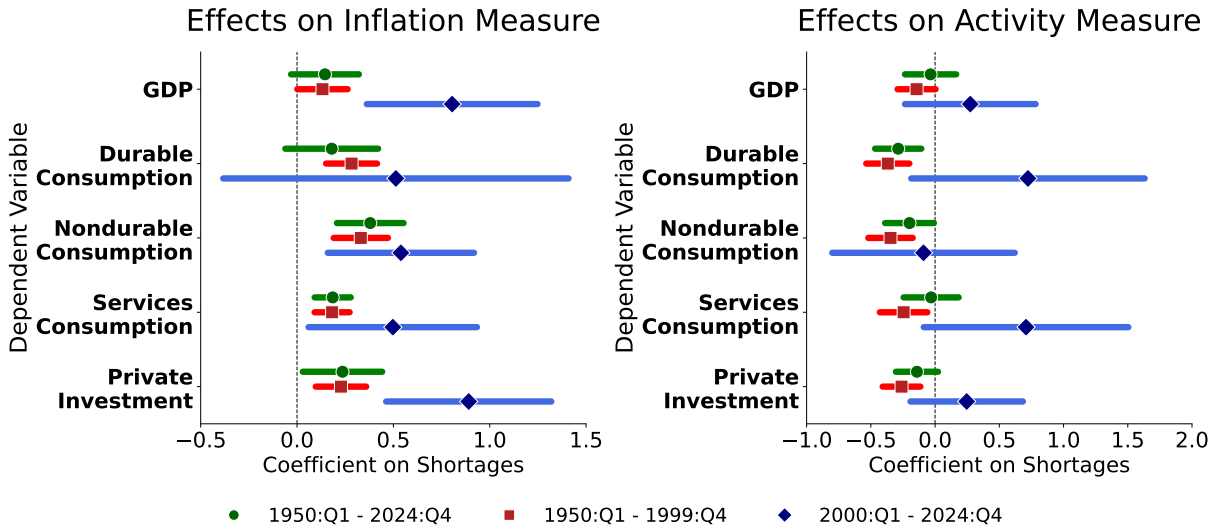
$$\Delta Y_{t+h} = \frac{400}{h} \ln\left(\frac{Y_{t+h}}{Y_t}\right)$$

represents the annualized log change of a variable of interest Y_t between period t and horizon h, and $SHORTAGE_t$ denotes the shortage index. The vector $\mathbf{X}$ contains control variables. For this exercise, we use quarterly BEA data from 1950:Q1 through 2024:Q4 for real GDP, personal consumption expenditures, and private fixed investment, each divided by the U.S. population to express it in per-capita terms. Inflation for each category is measured using the associated price deflator.

For each price and economic activity indicator, we estimate equation (1) by OLS using heteroskedasticity and autocorrelation consistent (HAC) standard errors following Newey and West (1987). As control variables, we include two lags of the shortage index and quarterly changes of the dependent variable and the corresponding economic indicator or price deflator, both contemporaneously and with two lags. For example, the regression for real GDP growth includes contemporaneous and lagged values of both real GDP growth and inflation (measured by the log change in the GDP deflator).

Figure 6 presents the results for four-quarter-ahead regressions (the Appendix reports results for the one- and eight-quarter horizons). For ease of comparison, we report standardized estimates of the coefficient β. In the full sample, higher shortages are associated with a

Figure 6: Relationship between Shortages and Inflation and Activity 4-Quarters Ahead



Note: Standardized coefficients from quarterly predictive regressions of inflation and activity on the shortage index. The dependent variable is the log difference between $t+4$ and t of the variable in each row (the price deflator for inflation, real quantities for growth). Controls are the quarterly changes of the dependent variable and of the associated activity or price measure, contemporaneously and with two lags. The full sample runs from 1950:Q1 to 2024:Q4, with subsamples 1950:Q1–1999:Q4 and 2000:Q1–2024:Q4. Shaded bars are heteroskedasticity-consistent 90 percent confidence intervals.

rise in inflation across GDP components. The effects on economic activity differ somewhat across components: consumption of both durable and nondurable goods and private investment decline significantly, while services consumption and aggregate GDP show small and statistically insignificant effects—differences that, as we show below, largely reflect time variation. The standardized coefficients imply that, for instance, a one-standard-deviation increase in the shortage index is associated with a rise in durable goods inflation of about half a percentage point and a decline in durable goods growth of about 2 percentage points.

Figure 6 also documents substantial time variation in these relationships. When we split the sample into two periods, pre- and post-2000, selecting the cutoff so that each subsample has enough observations for estimation, shortages exhibit a positive correlation with future inflation across components throughout. Their link with economic activity shifts from significantly negative in the earlier period to significantly positive in the latter. Within each subsample, the estimates for both inflation and activity are similar across GDP components. Thus, the weak and insignificant full-sample coefficients on services and aggregate GDP reflect averaging across periods with offsetting signs.

A further concern is whether the shortage index contains information beyond that captured by other macroeconomic variables. We address this issue by augmenting our baseline regression with additional controls, including oil prices, commodity prices, wage growth, and inflation expectations. The inclusion of these controls only slightly attenuates the effect of shortages on inflation and activity, but the coefficients remain in large part statistically significant (see the Appendix for details).

4.2 Time Variation in Shortage Effects: Rolling Regressions

To assess the stability of our findings across different historical periods, we estimate equation (1) using rolling 30-year backward-looking windows. Each point on the chart at date t shows the coefficient estimated using data from the preceding 30 years. For example, the coefficient shown for 1973 is based on data from 1943 through 1973. We focus on real GDP growth and its deflator, extending the sample back to 1900:Q1.[15]

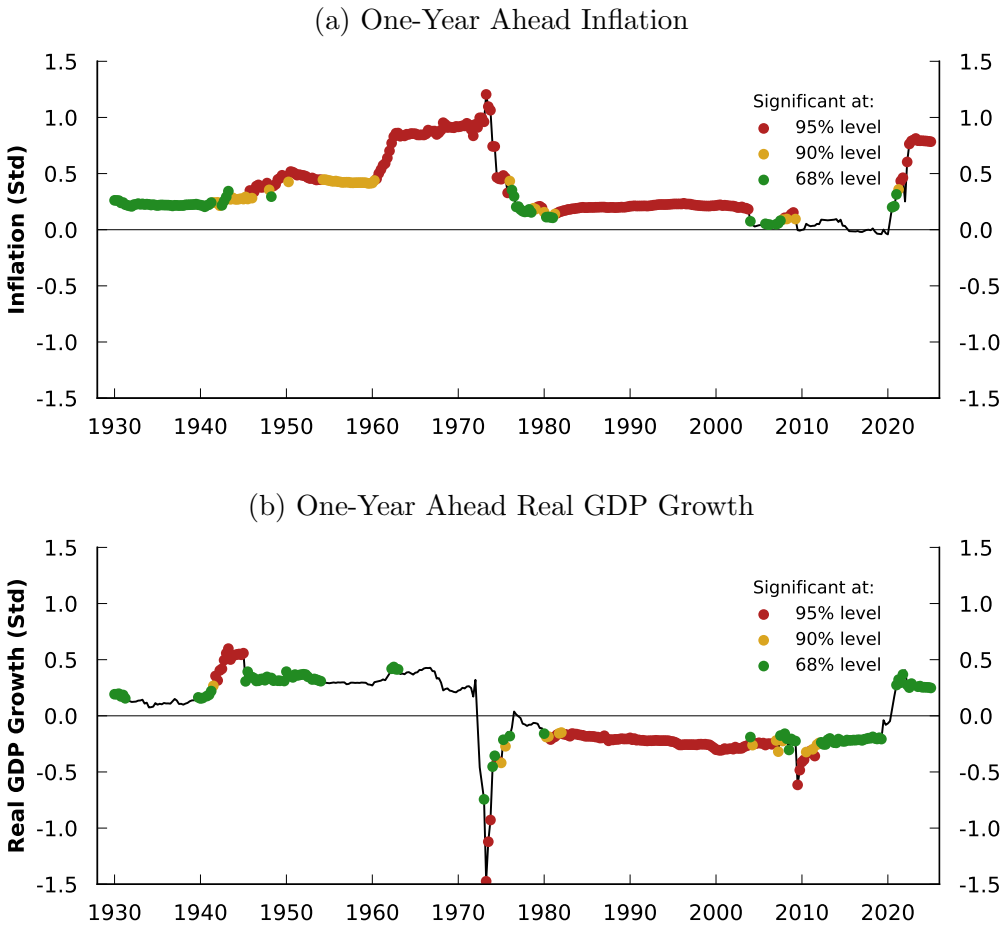
Figure 7 reveals stability in the shortage-inflation relationship alongside substantial variation in the shortage-activity relationship. Shortages consistently predict higher inflation starting in the 1940s, with coefficients remaining positive and statistically significant across nearly all rolling windows. The effect becomes negligible only during the decade preceding the pandemic, when shortage episodes were rare. This positive relationship holds across diverse historical episodes—including both world wars, the 1970s oil shocks, and the post-pandemic period.

The relationship with economic activity, by contrast, exhibits marked time variation. Coefficients are positive but generally insignificant in rolling windows through the 1960s, with the exception of periods when World War II—an episode with large positive and significant effects—is included in the sample. Once World War II exits the 30-year window, coefficients turn significantly negative, remaining so from the early 1970s through the early 2000s. In the most recent period, coefficients become positive and significant again—though this reversal reflects the COVID-19 pandemic quarters within the window dominating the estimate rather than a broad change across the full window.

All told, the rolling estimates on inflation and activity suggest that the drivers of shortage episodes differ across historical periods. During World War II and the COVID-19 pandemic, shortages coincided with strong demand pressing against binding supply constraints. In

[15] We extend the sample for GDP, the GDP deflator, and population using data from Ramey and Zubairy (2018).

Figure 7: Effect of Shortages on Inflation and Real GDP Growth (30-Year Window)



Note: The figures show the estimated effect of shortages on inflation and real GDP growth from regressions over rolling 30-year windows, using data back to 1900. The dependent variable is the 4-quarter-ahead difference in the log GDP deflator or in log real GDP per capita; the main regressor is the shortage index. Controls are the 1-quarter changes in both variables in quarter t and two lags. Significance uses heteroskedasticity-consistent standard errors.

contrast, from the early 1970s through the early 2000s they arose primarily from supply disruptions and market frictions that constrained activity.

4.3 Predictive Regressions with Categorical Shortage Indexes

Although the four categorical shortage sub-indexes are correlated with one another, each captures distinct, sector-specific variation. We test this directly by regressing selected price and wage indicators on the four categorical shortage indexes using monthly data from 1964:M1

Table 3: Predictive Regressions with Categorical Shortage Indexes

	Wages		PPI Materials		CPI Energy		CPI Food	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Labor Shortages	0.06*	0.01		-0.18		0.01		-0.14
	(1.67)	(0.12)		(-1.54)		(0.10)		(-1.53)
Industry Shortages		0.05	0.25***	0.43***		0.16		0.34***
		(0.34)	(2.86)	(3.21)		(1.63)		(3.10)
Energy Shortages		0.16		0.14**	0.26***	0.21***		0.04
		(1.50)		(2.08)	(5.44)	(2.99)		(0.61)
Food Shortages		-0.01		-0.16**		-0.09	0.19***	0.07
		(-0.23)		(-2.43)		(-1.20)	(3.88)	(1.37)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	725	725	729	729	729	729	729	729

Note: The table reports standardized coefficients of predictive monthly regressions of selected prices and wages indicators on the categorical shortage indexes. The dependent variable for each regression is the 3-month log difference of the producer price index (PPI) for processed goods for intermediate demand, consumer price index (CPI) energy, and CPI Food, and the 12-month log change in average hourly earnings (total private). Each regression includes as controls three lags of the monthly changes of the dependent variable and one lag of the log change in industrial production. The sample runs from 1964:M1 through 2024:M12. Heteroskedasticity and autocorrelation consistent t-statistics are reported in parentheses and computed according to Newey and West (1987). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

through 2024:M12. The dependent variables are the price and wage measures most exposed to shortages in each category: the 3-month log difference of the producer price index (PPI) for intermediate materials, the CPI for energy, the CPI for food, and the change in average hourly earnings.[16] Each regression includes three lags of the monthly change in the dependent variable and one lag of the log change in industrial production.

When included on its own, each categorical index forecasts the prices most exposed to its own sector. Table 3 shows that these results largely survive when all four indexes are included simultaneously, with one exception: food shortages no longer predict CPI food inflation once the other categories are controlled for. Because each sectoral index predicts the prices and wages most exposed to that sector's shortages, the disaggregated components carry sector-specific information, not a common measure of news volume.

The results in this section show that shortages contain predictive information for both

[16] We use the 12-month change for earnings as they adjust more slowly than commodity prices or materials.

inflation and economic activity—at the aggregate and sectoral level alike—with effects that vary across historical episodes. The next section uses a structural model to separate shortage shocks from conventional demand and supply shocks.

5 Shortages: A VAR Analysis

In this section, we estimate a structural vector autoregression (VAR) in inflation, activity, and shortages. The model's key departure from standard models of aggregate demand and supply is a wedge between desired demand and realized output driven by shortages. We use the estimated model to measure how strongly shortages respond to demand and supply shocks, to trace the effects of shortage shocks, and to decompose major historical episodes, including the post-pandemic inflation surge. Following Baumeister and Hamilton (2019), we identify the model with informative priors on its structural parameters rather than exact exclusion restrictions.[17]

5.1 The Model

The VAR model includes three quarterly variables: inflation, activity, and shortages. These are summarized by the vector:

$$\mathbf{X}_t = (\pi_t, y_t, h_t)' \tag{2}$$

where inflation (π_t) and activity (y_t) are quarter-over-quarter percent changes in the U.S. headline CPI and real GDP. Shortages (h_t) are measured by our shortage index.[18] All series are expressed in deviation from their sample mean, and our shortage index is standardized, so that values below (above) zero indicate lower-than-average (higher-than-average) shortages. A below-average reading means shortage news is quieter than usual, not that the economy is in a state of slack: the index is built to detect shortages, not their absence. The sample runs from 1950:Q1 through 2024:Q4.

The system of equations consists of an aggregate supply equation, an aggregate demand equation, and an equation connecting shortages with aggregate demand and aggregate supply.

[17] An alternative approach would be a VAR with nonlinearities or time-varying parameters. Given the lack of consensus on the form such nonlinearities should take, we view a linear structural VAR as a natural first step.

[18] We deliberately maintain a parsimonious three-variable specification, which keeps the shortage wedge transparent and avoids the curse of dimensionality. In the Appendix we expand the model to include the interest rate.

The structural representation is given by:

$$\mathbf{A}_0\mathbf{X}_t = \sum_{j=1}^{p} \mathbf{B}_j\mathbf{X}_{t-j} + \mathbf{u}_t, \tag{3}$$

where $\mathbf{u}_t = (u_t^S, u_t^D, u_t^H)'$ is the vector of structural shocks—supply, demand, and shortage shocks—with zero mean and a diagonal covariance matrix $E[\mathbf{u}_t\mathbf{u}_t'] = \mathbf{\Sigma}_u$.

The structural relationships among variables, with $\mathbf{z}_{t-1} = (\mathbf{X}_{t-1}', \mathbf{X}_{t-2}', \ldots, \mathbf{X}_{t-p}')'$ denoting the vector of lagged endogenous variables, are described by the following equations:

$$\pi_t = \kappa\ y_t + [\mathbf{b}^S]'\mathbf{z}_{t-1} + u_t^S, \tag{4}$$

$$y_t + \alpha\ h_t = -\delta\ \pi_t + [\mathbf{b}^D]'\mathbf{z}_{t-1} + u_t^D, \tag{5}$$

$$h_t = \theta_S\ u_t^S + \theta_D\ u_t^D + [\mathbf{b}^H]'\mathbf{z}_{t-1} + u_t^H. \tag{6}$$

Equation (4) represents the economy-wide aggregate supply relationship, where inflation is positively related ($\kappa > 0$) to realized activity y_t and subject to supply shocks u_t^S.

Equation (5) characterizes the aggregate demand relationship, augmented to incorporate a shortage wedge. As in Baumeister and Hamilton (2018), desired demand d_t is assumed to depend negatively on inflation with an elasticity of δ. We assume that shortages create a gap between desired demand and realized activity y_t, so that desired demand can be expressed as $d_t \equiv y_t + \alpha h_t$, where $\alpha > 0$ is a scaling parameter converting the shortage index into units of economic activity. The shortage wedge αh_t thus quantifies unmet demand—the difference between what economic agents wish to purchase and what actually gets produced and sold—relative to its average level, since h_t is demeaned. This gap arises when frictions prevent prices from adjusting sufficiently to equate supply and demand. Because h_t enters the demand equation, shocks that raise shortages lower realized activity for a given level of desired demand.

Equation (6) models the shortage index. Traditional approaches to modeling demand and supply in multivariate models have no such wedge: implicitly, $h = 0$ at all times and production always coincides with desired demand. Here we model shortages as a direct function of the structural shocks hitting the economy. Shortages increase when demand surges ($\theta_D > 0$) or when supply is disrupted ($\theta_S > 0$), to the extent that prices do not fully adjust.

A shortage requires two things at once: a disturbance that pushes demand above supply at

the prevailing price, and a friction that keeps the price from closing the gap. The parameters θ_S and θ_D capture the shortages that supply and demand shocks generate under typical frictions. The shortage equation also includes an exogenous structural shock u_t^H, which captures shortages beyond what these shocks would normally produce, because frictions bind unusually tightly or because the allocation of available supply breaks down. As Section 2 documents, the friction most strongly associated with shortages in our data is price controls, which co-occur with shortages far more often than any other phenomenon we examine. Other frictions, such as coordination failures or breakdowns in distribution networks, can also play a role.

Major economic events therefore typically involve a combination of shocks in our framework. Wartime episodes, for instance, combine genuine reductions in productive capacity (supply shocks u_t^S) with price controls and distribution disruptions that make the resulting shortages far larger than usual (shortage shocks u_t^H).

5.2 Identification and Estimation

We treat identification and estimation of this structural VAR model as a Bayesian exercise, imposing priors directly on the structural parameters $(\kappa, \alpha, \delta, \theta_S, \theta_D)$ and on the lag coefficients in $[\mathbf{b}^S]'$, $[\mathbf{b}^D]'$, and $[\mathbf{b}^H]'$. Priors on the contemporaneous structural parameters discipline the identification of the structural shocks, while priors on the lag coefficients regularize the dynamics of the system. All prior densities are intentionally wide, so that the data can update both their location and spread. We estimate the model with $p = 2$ lags; a more flexible specification with 8 lags has negligible effects on our main results (Figure 10). As a further robustness check, Appendix D augments the model with the policy rate to control for monetary policy; our main results are unchanged.

The priors for κ and δ in equations (4) and (5) follow inverse gamma distributions with a mean of 0.2 and a standard deviation of 1. At these values, a 1 percent demand-driven increase in activity raises inflation by 0.2 percentage points, and a 1 percent supply-driven increase in inflation reduces GDP growth by 0.2 percentage points. These assumptions are broadly consistent with estimates from both DSGE models (Smets and Wouters, 2007) and structural VAR models of the U.S. economy (Baumeister and Hamilton, 2018). The shortage response parameters θ_S and θ_D share inverse gamma priors with a mean of 0.2 and a standard deviation of 1. These priors reflect the notion that an expansionary supply shock (like a productivity boom) should relax capacity constraints, whereas a positive demand shock should tighten them.

For the shortage scaling parameter α, we assign a beta prior with mean 0.25 and standard deviation 0.125, a value that implies modest but meaningful shortage effects on the activity gap, while allowing substantial updating by the data.[19]

We center the priors on the lag coefficients and on the standard deviations of the structural shocks on an auxiliary VAR evaluated at the prior means of the structural parameters, shrinking the cross-lag coefficients toward zero (Appendix D). The one exception is the coefficient on lagged shortages in the inflation equation, $\rho_{1,\pi H}$, whose prior mean we set at 0.10, so that a shortage shock leaves inflation unchanged in the first year on average while letting the data speak regarding its overall inflationary effect. Our inverse-gamma prior on the shock volatilities also implies Student-t marginal distributions for the structural innovations, so the model accommodates the large, infrequent shocks that characterize episodes such as the world wars, the oil crises, and the pandemic. Priors and posteriors for the key structural parameters are shown in Appendix Figure A.7.

5.3 Results

We estimate the model using the algorithms in Dynare (Adjemian et al., 2011). The first 20 observations serve as a training sample, and we draw posterior samples using a random walk Metropolis-Hastings algorithm.

Parameter Estimates

Appendix Figure A.7 shows the prior and posterior distributions of the key parameters.[20] The posteriors are substantially updated relative to the priors, indicating that the data are informative about the structural parameters.

The posterior mean of the shortage wedge parameter α is 0.18: when the shortage index is one standard deviation above its mean, realized output falls short of desired demand by approximately 0.18 percentage points. The posterior means of κ and δ imply a flatter supply curve and a steeper demand curve than the prior assumed, so inflation responds less to shifts

[19] Because h is standardized, capping α at 1 imposes that a single standard deviation increase in shortages cannot open a demand-supply wedge larger than 1 percentage point of GDP. An upper bound of 1 for a single standard deviation is highly permissive, and prevents the sampler from wandering into economically absurd regions where a mild shortage event creates a large gap between demand and supply.

[20] Appendix Table A.2 shows the prior distributions for all model parameters, along with their posterior mean, standard deviation, and 80 percent credible intervals.

in demand and more to shifts in supply. Shortages respond to both demand and supply shocks, but about three times more strongly to supply shocks (θ_S versus θ_D). Finally, the data push the coefficient on lagged shortages in the inflation equation, $\rho_{1,\pi H}$, above its prior mean, with nearly all of its posterior mass in the positive region: after controlling for current and lagged movements in all other variables, higher shortages raise inflation in the following quarter.

Impulse Responses: Shortage versus Supply Shocks

Figure 8 reports the impulse responses. The posterior intervals are narrower than those implied by the prior, indicating that the data are informative about these responses. A shortage shock leads to a persistent rise in inflation and a slow, persistent decline in GDP. Shortages also respond to other shocks, rising after an expansion in aggregate demand or a contraction in aggregate supply. Shortage shocks thus move activity and inflation, while demand and supply shocks move shortages.

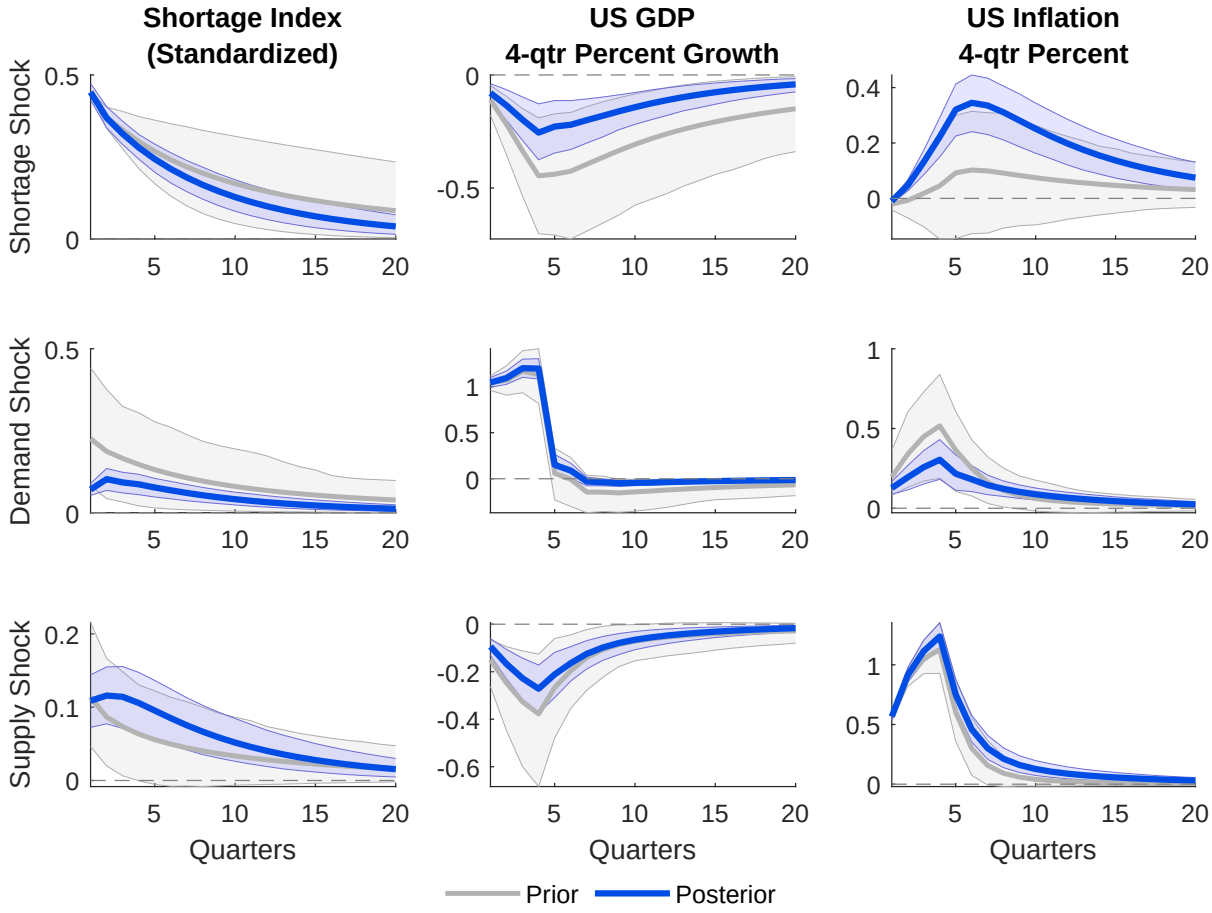
Two features distinguish shortage shocks from an ordinary supply shock. First, their inflationary effect is more persistent. Following a supply shock, four-quarter inflation peaks within a year and then falls quickly, with a half-life of about one and a half quarters. Following a shortage shock, inflation builds for about a year and a half before peaking, and takes about two years to decay by half. This delayed and persistent response, which reflects the effect of lagged shortages on inflation, makes the shortage index a leading indicator of inflation and GDP.

Second, shortage shocks produce a different inflation-output trade-off. Comparing peak responses at the posterior mean, a shortage shock lowers four-quarter GDP growth by about 0.74 percentage points for each percentage point of four-quarter inflation, against 0.23 for a supply shock—roughly three times as much output lost per unit of inflation. Shortage shocks act as demand wedges that constrain activity, so quantities, not just prices, absorb the disruption.

The Effects of Shortages throughout History

Figure 9 presents the historical decomposition, showing how the three structural shocks have contributed to movements in shortages, GDP growth, and inflation. Shortage shocks are the main driver of the shortage index, but a modest driver of GDP and inflation, and they typically operate together with traditional shocks. Demand and supply shocks also contribute

Figure 8: Effects of Shortages on US Activity and Inflation in the Structural VAR Model



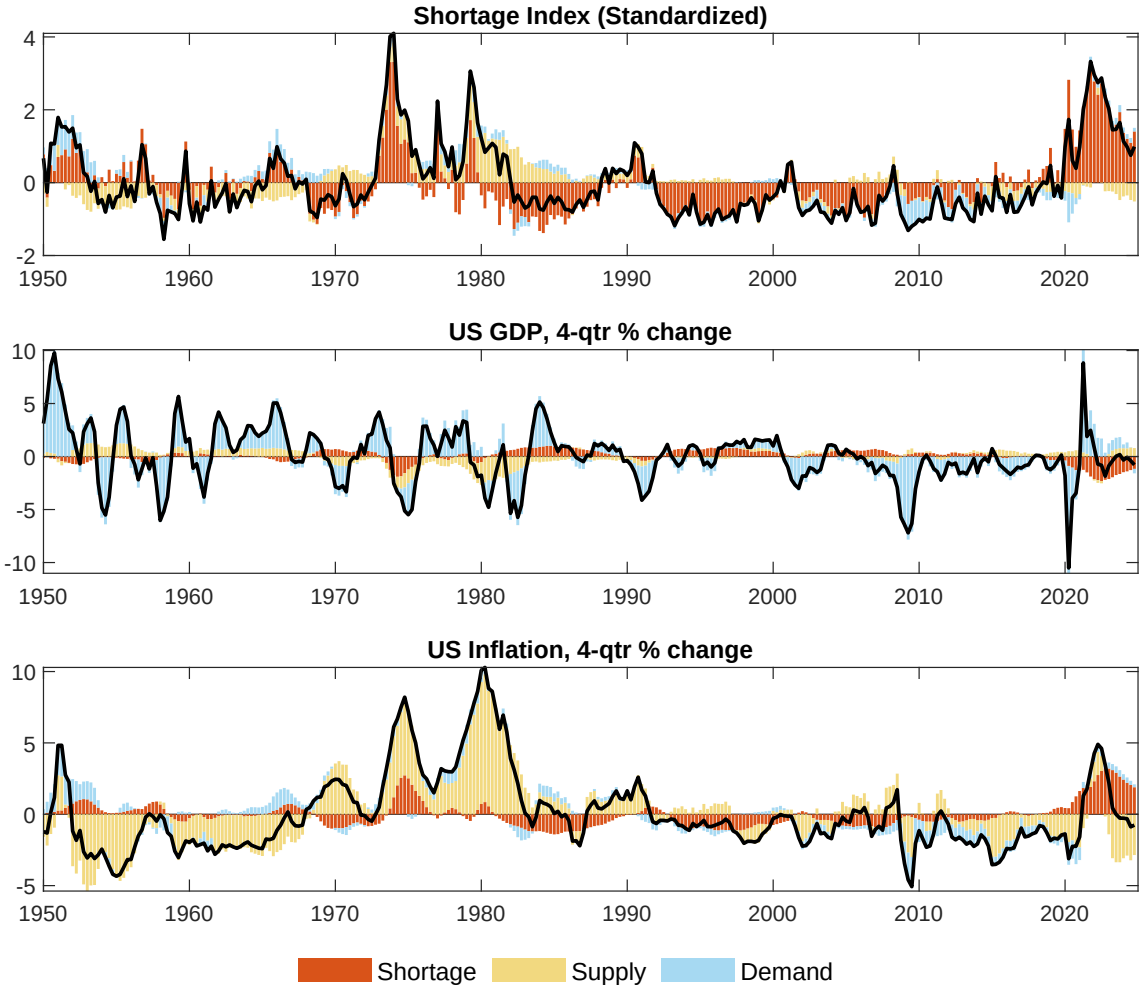
Note: Impulse responses to one-standard-deviation structural shocks in the VAR model of Section 5. Gray and blue lines are the responses at the prior and posterior mean; shaded areas are the corresponding 80 percent confidence intervals.

to shortages around key episodes. In the early 1950s, shortages reflect both demand shocks from the transition to a peacetime economy and shortage-specific disruptions related to the Korean War. In the 1970s, the oil embargo appears as a traditional supply shock, but the period also features orthogonal shortage shocks—reflecting the era's price controls and allocation rules—that contributed independently to higher inflation and slower growth.

Shortages, Supply, and Demand in the Post-Pandemic Inflation Surge

Much of the debate about the post-pandemic surge in inflation has been framed in terms of demand versus supply: Giannone and Primiceri (2024) and Bergholt et al. (2026) find that demand was the main driver, while Ascari et al. (2024), Comin et al. (2023), and Shapiro

Figure 9: Historical Decomposition from the VAR: Full Sample

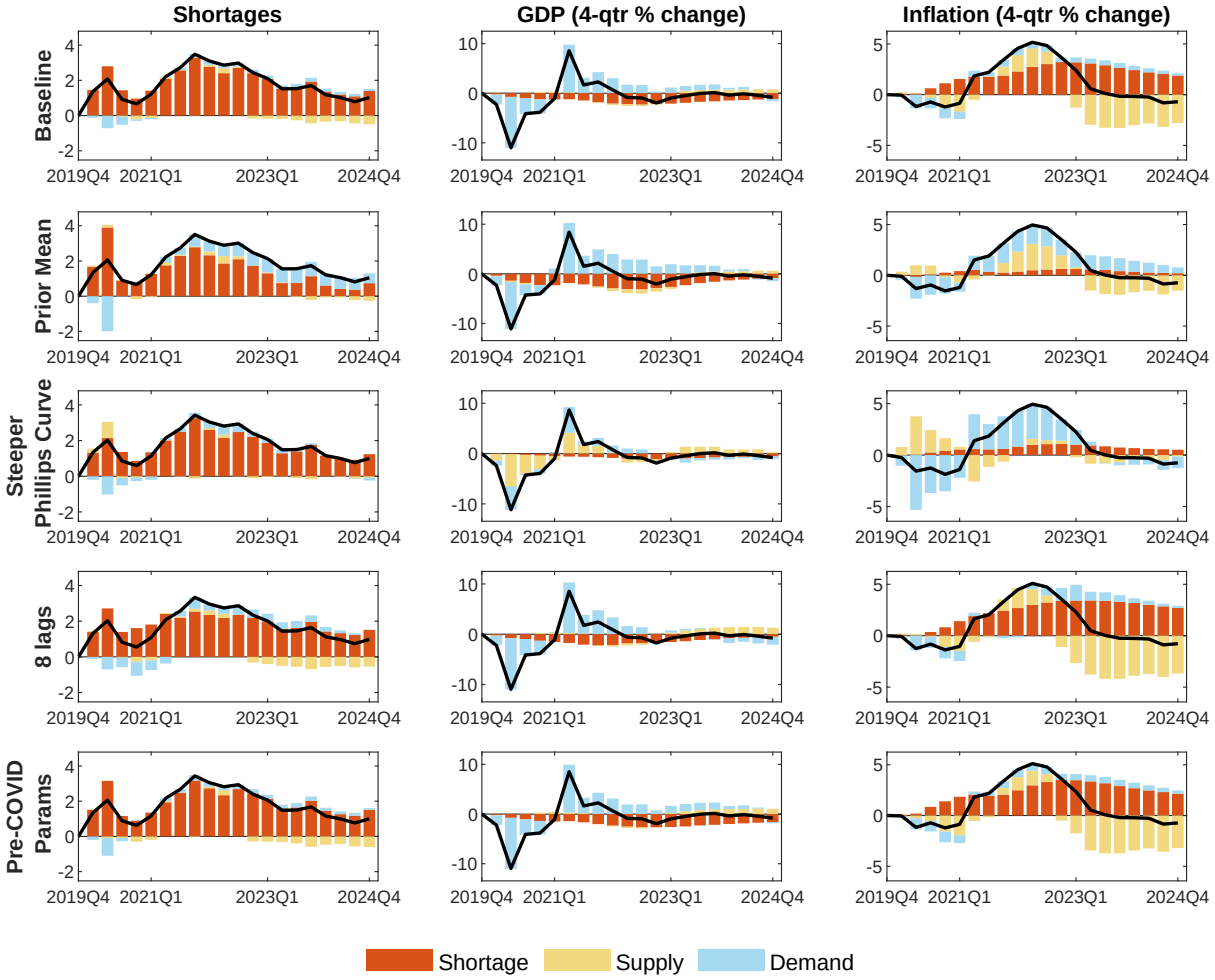


Note: Historical contributions of the structural shocks to the model's endogenous variables, expressed as deviations from their sample means.

(2026) emphasize a significant role for supply. Our model adds a third source, shortage shocks.

In our baseline model (top row of Figure 10), the surge was driven mainly by supply forces—traditional supply shocks and shortage shocks—with demand playing a secondary role. In 2020:Q1–2021:Q1, adverse shortage shocks—labor market disruptions and supply chain bottlenecks—coincide with depressed demand, so shortages rise but inflation stays close to baseline. In 2021:Q2–2022:Q3, all three shocks contribute: shortage shocks, reflecting the rebalancing of demand from services to goods, port congestion, and semiconductor bottlenecks, continue to depress activity; supply shocks push up inflation; and strengthening demand adds inflationary pressure. From 2022:Q4 onward, shortages decline but remain above pre-pandemic levels, and given their delayed effects, past shortages continue to raise inflation well into 2024.

Figure 10: Historical Decomposition from the VAR: 2020-2024, Alternative Models



Note: Contributions of the structural shocks since 2020, in deviation from 2019:Q4; shocks before 2019:Q4 are set to zero. Rows: baseline at the posterior mean; all parameters except shock volatilities at the prior mean; steeper Phillips curve ($\kappa = 1$); eight lags; and pre-COVID parameters (estimated through 2019:Q4, then held fixed to filter the 2020–2024 shocks).

The contribution of shortage shocks is informed by the data, not imposed by our priors. With all parameters except the shock volatilities held at their prior means (second row), shortage shocks account for almost none of the surge, and the model attributes inflation to traditional supply and demand shocks. At the posterior mean, shortage shocks become meaningful contributors.

Whether the surge is attributed to supply or to demand hinges on the estimated slopes of the supply and demand curves. When we restrict the supply curve to be steep ($\kappa = 1$, third row), the model estimates a flatter demand curve, and demand shocks become the

dominant driver, consistent with Giannone and Primiceri (2024). Models that emphasize demand pressures thus feature relatively flat demand curves, whereas those that emphasize supply constraints require steeper ones. Within our framework, the data favor the baseline specification, which has a higher marginal likelihood.

Two further exercises support the baseline reading. First, a specification with 8 lags instead of 2 (fourth row) yields a nearly identical decomposition. Second, the pandemic itself may shape the parameters used to interpret it. The bottom row therefore reports a pre-COVID specification: we estimate the structural parameters on data through 2019:Q4 and then hold them fixed, filtering the 2020–2024 shocks through these pre-COVID parameters rather than re-estimating them over the pandemic period. The resulting decomposition is close to the baseline. The pre-COVID estimates imply a slightly flatter supply curve (a Phillips-curve slope of 0.105 versus 0.121 in the baseline), so they attribute a slightly larger share of the surge to supply and shortage shocks.

6 Conclusions

This paper introduces a novel monthly newspaper-based shortage index, a measure of the unmet demand that official statistics do not record, covering 1900 through 2025. The index tracks shortages across sectors—industrial goods, labor, food, and energy—and exhibits pronounced spikes during periods of economic turmoil, such as the world wars, the oil crises of the 1970s, and the COVID-19 pandemic. An audit against a large language model shows that the index reliably identifies shortages, and its historical coverage reaches well beyond existing measures of supply-chain pressure.

Predictive regressions show that increases in the shortage index predict persistently higher inflation throughout the past century, while their relationship with economic activity has varied over time. Structural VAR analysis shows that shortage shocks are not simply another supply shock: their inflationary effects are far more persistent, and for the same rise in inflation they generate larger output losses. The same forces are visible in the historical record: shortage shocks contributed to the 1970s stagflation and to the post-pandemic inflation surge, in both cases alongside—not instead of—traditional supply and demand shocks.

Macroeconomic models typically abstract from shortages or treat them as transitory symptoms of sluggish price adjustment. Our findings suggest otherwise: shortages recur

throughout economic history and have effects on inflation and activity of their own. Our analysis relies on linear models, whereas recent work implies that shortages may propagate nonlinearly—through production complementarities and binding capacity constraints (Baqaee and Farhi, 2022; Comin et al., 2023), or through rationing when prices do not clear markets (Holden, 2025; Flynn et al., 2026; Liu, 2026). Our index, spanning more than a century of U.S. history, provides the long time series needed to measure shortage episodes and to test these theories.

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Appendix for: Measuring Shortages since 1900

A Data Sources

This Section describes in detail the data used in the paper in addition to the shortage indexes presented throughout the paper and available at https://www.matteoiacoviello.com/shortages.html.

The shortage indexes presented in the paper are constructed and analyzed using search queries that run against two ProQuest newspaper databases, Global Newsstream Collection and Historical Newspapers, using ProQuest's TDM Studio text and data mining tool (see https://tdmstudio.proquest.com/home).

Section 4 describes the results of regressions of selected price and wage indicators on the categorical shortage indexes. In addition to the shortages indexes, the series we use in these regressions and the corresponding data sources are:

- Wages, measured by BLS Average Hourly Earnings in Total Private Industries (Haver mnemonics: LETPRIVA@USECON);
- Producer Prices, measured by PPI: Intermediate Demand Processed Goods, seasonally adjusted (Haver mnemonics: PC1@USECON);
- CPI: Energy (Haver mnemonics: PCUSEN@USECON);
- CPI: Food (Haver mnemonics: PCUFON@USECON).

Section 3 also compares our shortage index with alternative measures of supply constraints. Data sources are as follows:

- The Global Supply Chain Pressure Index (GSCPI) published by the Federal Reserve Bank of New York and described in Benigno et al. (2022) (Haver mnemonics: GSCPI@SURVEYS);
- The ISM Supplier Deliveries Index (SDI) produced by the Institute for Supply Management (Haver mnemonics: NAPMVDIN@SURVEYS);
- U.S. Supply Bottlenecks Index (SBI) from Burriel et al. (2024).

The predictive regressions in Section 4 use data on real GDP, personal consumption expenditures (durables, non-durables, and services), and private fixed investment and their associated deflators. Data for all variables cover the period 1950:Q1-2024:Q4 and are taken from the Bureau of Economic Analysis National Income and Product Account Historical Tables. Variables are expressed in per capita terms dividing by total population from the U.S. Census Bureau, taken from Haver (Total Population: All Ages including Armed Forces Overseas; Haver mnemonics: POPT@USECON). The population series starts in 1952 and is backcast to 1950 using population data from Ramey and Zubairy (2018).

In Section 4, we run the predictive regressions for GDP and its deflator on a rolling sample for the period 1900:Q1-2024:Q4. To this end, we backcast the GDP, GDP deflator, and population data described in the previous paragraph using the growth rates of the corresponding series in Ramey and Zubairy (2018), which are available for the earlier period.

The regressions in Section 4 also use as controls the following series:

- Oil Prices, measured by Spot Oil Price: West Texas Intermediate. (Haver mnemonics: PZTEXP@USECON);
- Commodity Prices, measured by the S&P Goldman Sachs Commodity Nearby Index (Haver mnemonics: GSCI@USECON);
- Oil Prices and Commodity Prices are expressed in real terms by dividing for the Seasonally Adjusted PCE Chain Price Index (Haver mnemonics: JCBM@USECON);
- Wages, measured by BLS Average Hourly Earnings in Total Private Industries from the BLS (Haver mnemonics: LETPRIVA@USECON);
- 10-year Inflation Expectations from the Federal Reserve Bank of Cleveland, retrieved from FRED (Fred mnemonics: EXPINF10YR).

The VAR analysis of Section 5 uses the following variables:

- U.S. real GDP, from the Bureau of Economic Analysis;
- CPI for All Items (Haver mnemonics: PCUN@USECON);

B Details of the Index Validation

The audit described in Section 3.1 uses Claude Opus 4.8 to establish ground truth on a random sample of 10,296 articles drawn from all of the index's constituent newspapers except *The Wall Street Journal*, for which full article text was not available for running the model. This appendix provides additional detail on that validation: the coded examples given to the model, the LDA methodology used to support the four-topic taxonomy, and a robustness check that corrects the index for the estimated time variation in its false discovery rate and miss rate.

Coded examples

Along with the definition and scoring rules, the classification prompt gives the model a fixed set of short worked examples that pin down the treatment of the hard cases. Several are deliberately scored zero even though they contain the word "shortage"—denials, figurative uses, and shortages that fall outside our four categories—so that the model handles negations and scope consistently rather than firing on the keyword alone. Table A.1 reproduces the full set.

Table A.1: Coded Examples Provided to the Classifier

Example phrase	Score	Rationale
"steel shortages would force a plant closing"	1	industrial material
"gasoline rationing as the oil shortage deepened"	1	energy
"reservoirs could run dry, the city's water is running out"	1	water (no "shortage" word)
"supplies of meat are dwindling, shelves nearly empty"	1	food/goods
"no shortage of cars this year"	0	denial
"no shortage of conspiracy theories"	0	not economic
"a shortage of workspace / of parking"	0	space/room, out of scope
"the city's housing shortage"	0	housing
"policemen wait as the complaint room stays backed up"	0	backlog, not supply
"the short list is at hand; time is running out for talks"	0	figurative
"a shortage of physicians / nurses / midwives"	0	medical labor
"an Ozempic / drug shortage left patients without medication"	0	drug, medical
"hospitals ran out of masks, gowns and gloves"	0	medical supplies (PPE)
"war-torn Europe faces food and coal shortages while its hospitals struggle"	1	in-scope food/energy present; health framing does not remove it
"a shortage of qualified appraisers / engineers / skilled machinists"	1	skilled/professional labor

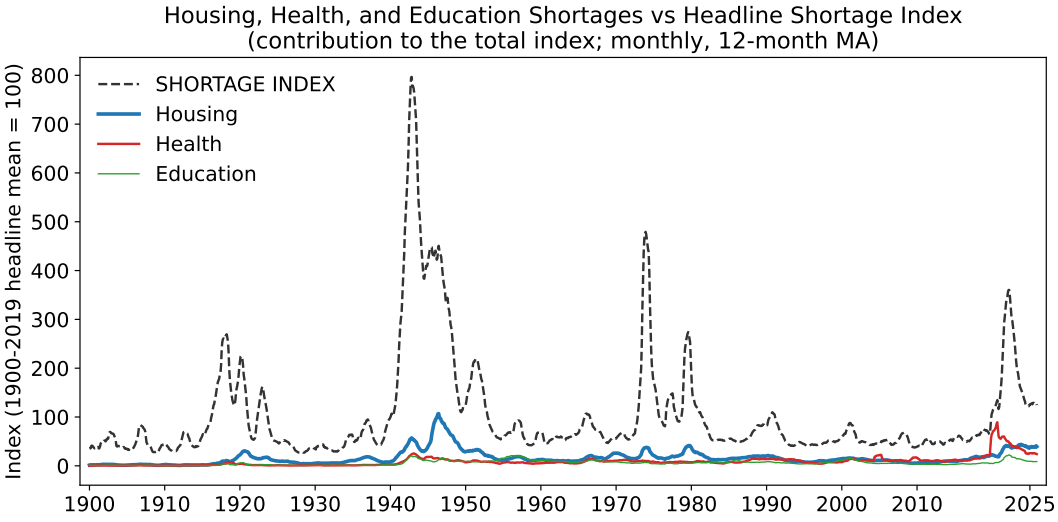
Note: The worked examples included in the classification prompt used to establish ground truth (Section 3.1). A score of 1 marks an economically relevant shortage of industry and goods, labor, food and water, or energy; a score of 0 marks anything outside these four categories, as well as denied or figurative uses. Examples scored 0 despite mentioning a "shortage" guard against false positives from negations and out-of-scope subjects.

LDA methodology

We fit the LDA model on the AI's own shortage explanation rather than on article text. For each article in the labeled sample (Section 3.1) that the AI flags as a shortage, we use the short free-text rationale it returns for that classification, so the model sees a concise, boilerplate-free statement of why the article was coded as a shortage rather than the whole article body. The classification prompt does not reference our energy/food/industry/labor taxonomy; the model is asked only to score the article and, separately, to justify that score in one sentence. We lowercase, remove stopwords, and drop the shortage terms themselves (so the topics are defined by their context), and we retain terms that occur at least a set number of times in the corpus.

Topics are estimated by Latent Dirichlet Allocation (Blei et al., 2003) under variational Bayes inference, with a fixed number of topics and a fixed random seed, and with both the document-topic and word-topic priors left to adapt during fitting rather than fixed in advance. The four estimated topics are named, after estimation, by the keyword category on which each places the most weight (energy on *fuel/gasoline*, food on *food*, industry on *materials*, labor on *labor*), assigned one-to-one so that each of the four names is used once. The fit recovers the same four topics—energy, food and agriculture, industrial goods, and labor markets—which

Figure A.1: Excluded Categories: Housing, Health, and Education Shortages



Note: Shortage indexes for housing, health, and education, expressed as contributions to the headline index: each is the share of articles reporting a shortage in that category, on the headline's scale (monthly, 12-month moving average, 1900–2019 headline mean equal to 100). The dashed line is the headline index.

we display as word clouds (Figure 1, bottom panel) and track as topic shares over time.

Excluded Categories: Housing, Health, and Education

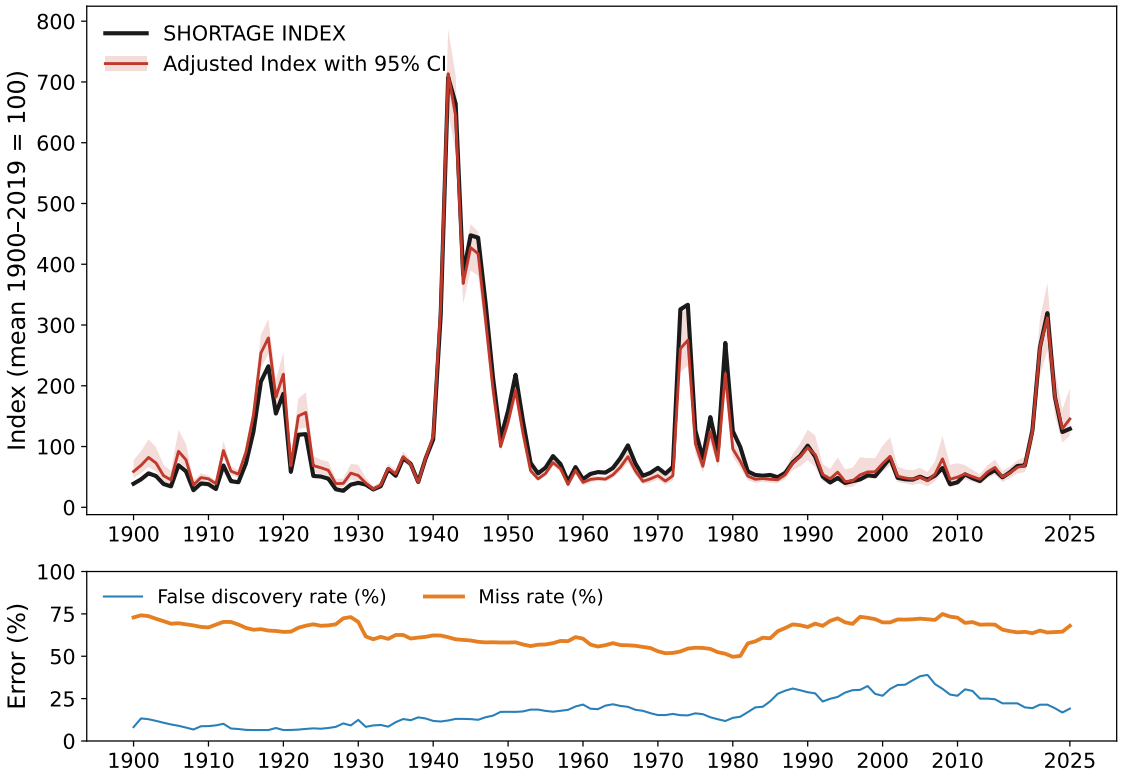
Figure A.1 plots the three excluded categories—housing, health, and education shortages—on the same re-indexed scale as the headline index (share of articles that are a shortage, monthly, 12-month moving average, normalized so the 1900–2019 mean of the headline equals 100). Because each category is scaled by the same constant as the headline, the curves show each category's contribution to the total, not a within-category share: a category sitting at 20 when the headline is at 100 accounts for one-fifth of the headline's level. The three series stay well below the headline throughout, with two exceptions—housing during the postwar shortage of 1946 (about a quarter of the headline) and health during the COVID-19 pandemic in 2020 (about two-thirds of the headline, in part reflecting pandemic-era medical vocabulary). Outside these episodes they add little, which is why we keep them out of the headline index.

Robustness: An Error-Adjusted Index

The frontier tells us the query's average false discovery rate and miss rate over the whole sample, but both rates drift over the century as the vocabulary of shortages changes. As a robustness check, we ask whether correcting the index for this time-varying measurement error would change its shape. It does not.

We first estimate how the two error rates evolve over time. We compute a moving average of the false discovery rate and the miss rate—the share of keyword hits that are false positives, and the share of true shortages the query misses—in a rolling window about twenty years

Figure A.2: The Error-Adjusted Shortage Index



Note: The top panel plots the raw shortage index (black) and the error-adjusted index (red) with a bootstrap 95 percent confidence band. The adjustment uses the estimated false discovery and miss rates in a precision/recall analogue of the Rogan and Gladen (1978) correction (see text). The bottom panel plots both rates, computed as an approximately 20-year moving average over the labeled sample.

wide.[1] This yields smoothly varying FDR_t and Miss_t series (bottom panel of Figure A.2): the miss rate declines fairly steadily from the early twentieth century to a low point around the 1970s, and the false discovery rate, after some fluctuation over the same span, rises together with the miss rate from the 1970s through the early 2000s before both ease somewhat over the most recent decades.

We then correct the raw index using these rates. Writing M_t for the measured (keyword) index, the error-adjusted index is

$$\widehat{S}_t \;=\; M_t \,\frac{1-\mathrm{FDR}_t}{1-\mathrm{Miss}_t},$$

the natural correction for misclassification: multiplying by $(1-\mathrm{FDR}_t)$, one minus the false-positive share of the hits, strips the false positives out of the measured count, and dividing by $(1-\mathrm{Miss}_t)$, the recall, grosses the count back up for the shortages the query missed. Because

[1] The window is defined by article count rather than calendar time, which keeps the number of labeled shortages behind each estimate roughly constant.

our two error rates are defined relative to the query's hits ($1 - \text{FDR}_t$ is precision) and to the true shortages ($1 - \text{Miss}_t$ is recall), the correction takes this multiplicative form, the false-discovery-rate/miss-rate analogue of the balanced probability analysis of Norris and Kahn (2006).[2]

The two error-rate series, however, are themselves estimated over the same labeled sample used to build the frontier in Section 3.1, not over the full corpus underlying the keyword index—and this is not a fixable limitation: if the true error rates were known exactly over the full corpus, we would already have full-corpus ground-truth labels, making the keyword index itself unnecessary. The error-adjusted index should therefore be read as illustrative of the likely direction and rough magnitude of the correction, not as a precise re-estimate of the true series.

Figure A.2 plots the two series. Because the miss rate is higher early in the sample, the correction tilts the adjusted index slightly above the raw index in that period—consistent with articles that describe a lack of basic goods without using words like "shortage" or "scarcity" being somewhat more common in the earlier, more idiosyncratically worded press. A 1904 profile of a small Chicago bakery, for instance, reports that its owner's "only trouble is that she cannot supply the demand for her goods"—an unmistakable case of demand outrunning supply that never uses any of our search terms and so is missed entirely by the benchmark query. Text quality may also play a role. The text of older articles in ProQuest's Historical Newspapers comes from optical character recognition of scanned pages, and it is occasionally partly garbled, a known problem with digitized historical newspapers (Dell et al., 2023).[3] A garbled shortage word cannot match our query, while the model may still infer the shortage from the surrounding words.

The false discovery rate also rises from the 1970s through the early 2000s before falling back over the last decade and a half—moving together with the miss rate over this period (the two series have a correlation of 0.86 after 1970) rather than trading off against each other as the cross-sectional frontier in Figure 5a might suggest. This co-movement mostly reflects our classification scope, not a change in how journalists write: the benchmark query fires on "shortage," "scarcity," "bottleneck," and "rationing" wherever they appear, but Section 3.1's definition of a true shortage excludes several categories in which these words are common—housing, education, and government finance among them—as well as any hypothetical, historical, or explicitly denied shortage. A 1992 newspaper retrospective on Friedrich Hayek, for example, uses his own illustration of "a case like that of scarcity of one raw material" to explain how price signals coordinate economizing behavior—a purely theoretical device, not a reported shortage, and one the AI ground truth correctly excludes but our keyword query cannot distinguish from an actual report. As post-war press coverage

[2] This is closely related to the classic Rogan and Gladen (1978) misclassification correction, which is stated in terms of sensitivity and specificity; here we use the equivalent precision/recall form implied by our definitions of the two error rates. Norris and Kahn (2006) derive the mathematically equivalent relationship in the context of gene-expression significance lists: writing their false negative rate as $\text{FNR} = (\text{TTP} - \#\text{SL}\,(1 - \text{FDR}))/\text{TTP}$, where #SL is the number of genes on the significance list (our M_t) and TTP is the total number of true positives (our $\widehat{S}_t$), and solving for TTP gives exactly $\text{TTP} = \#\text{SL}\,(1 - \text{FDR})/(1 - \text{FNR})$, the same formula we apply to the shortage index. We bootstrap the labeled sample to place a 95% confidence band on $\widehat{S}_t$ (shaded in Figure A.2).

[3] For example, a 1923 *Washington Post* article on a Shriners convention opens "The pied piper would have to look to his laurels if he entered a contest with T. H. O'Callaghan," which the database text renders as "'fho piper hase to look to hia laurels tf he entered a ith T. 1I, O'Callaghan."

increasingly reached for "shortage"/"scarcity" language in these out-of-scope or figurative senses, both error rates rose together; the recent decline in both suggests this drift is not one-directional.

The reason is that the drift in the error rates is small relative to the movements in the index: the correction factor $(1 - \text{FDR}_t)/(1 - \text{Miss}_t)$ varies by about a factor of two over the century, whereas shortage episodes move the raw index by an order of magnitude. As a result, the index's profile is essentially unchanged: every major episode (the two world wars, the 1970s energy and labor shortages, the 2021–2022 supply-chain crunch) appears in the same place with the same relative magnitude. The two series are very highly correlated, both in levels and in annual changes—the object of interest for a business-cycle indicator. Correcting for time-varying classification error thus tilts the index modestly without altering the signal it carries.

C Additional Empirical Results

Additional Results from Predictive Regressions

We explore the relationship between shortages, inflation, and economic activity using the following predictive regression:

$$\Delta Y_{t+h} = \alpha + \beta\, SHORTAGE_t + \sum_{i=0}^{p} \boldsymbol{\gamma}_i' \mathbf{X}_{t-i} + \varepsilon_{t+h} \tag{7}$$

where

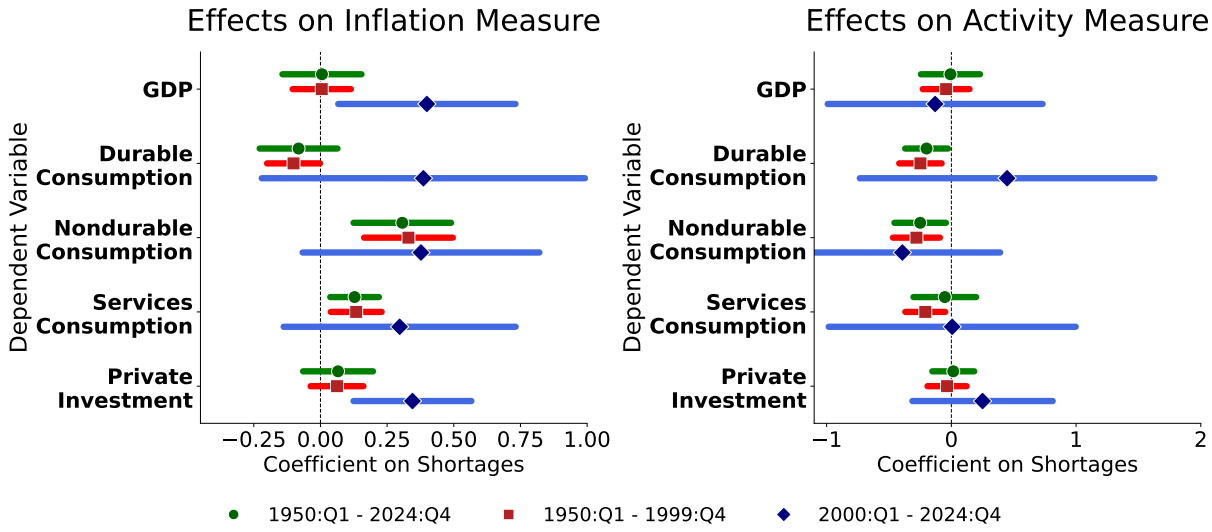
$$\Delta Y_{t+h} = \frac{400}{h} \ln\left(\frac{Y_{t+h}}{Y_t}\right)$$

represents the annualized log change of a variable of interest Y_t between period t and forecast horizon h, and $SHORTAGE_t$ denotes the level of the shortage index. The vector $\mathbf{X}$ contains control variables. We use quarterly data from 1950 through 2024 from the National Income and Product Accounts (NIPA) for real GDP, personal consumption expenditures, and private fixed investment, each divided by total population to express it in per-capita terms. Inflation for each category is measured using the associated price deflator. For GDP and its price deflator, we extend the sample back to 1900 using data from Ramey and Zubairy (2018).

For each price and economic activity indicator, we estimate equation (7) by ordinary least squares, with standard errors calculated following Newey and West (1987). As control variables, we include quarterly changes of the dependent variable and the corresponding economic indicator or price deflator, both contemporaneously and with two lags. For example, the regression for real GDP growth includes contemporaneous and lagged values of both real GDP growth and inflation (measured by the log change in the GDP deflator).

Figures A.3 and A.4 report results for the one- and eight-quarter horizons, respectively. For ease of comparison, we report standardized estimates of the coefficient β. A standardized coefficient represents the change in standard deviation units in the dependent variable following a one-standard-deviation change in the explanatory variable.

Figure A.3: Relationship between Shortages and Inflation and Activity One-Quarter Ahead



Note: Standardized coefficients from quarterly predictive regressions of inflation and activity on the shortage index. The dependent variable is the log difference between $t+1$ and t of the variable in each row (the price deflator for inflation, real quantities for growth). Controls are the quarterly changes of the dependent variable and of the associated activity or price measure, contemporaneously and with two lags. The full sample runs from 1950:Q1 to 2024:Q4, with subsamples 1950:Q1–1999:Q4 and 2000:Q1–2024:Q4. Shaded bars are heteroskedasticity-consistent 90 percent confidence intervals.

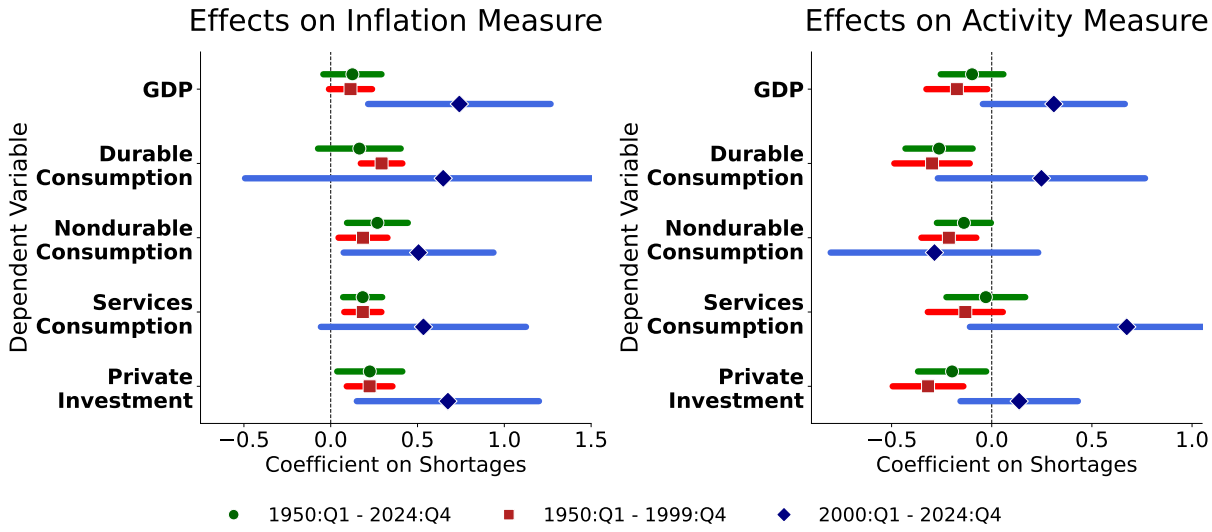
Does the Shortage Index Contain Information Beyond Traditional Indicators?

One concern discussed in the paper is whether the shortage index contains information beyond that captured by other macroeconomic variables. We address this issue by augmenting our baseline regression with additional controls, including oil prices, commodity prices, wage growth, and inflation expectations. The inclusion of these controls only slightly attenuates the effect of shortages on inflation and activity, but the coefficients remain in large part statistically significant. See Figure A.5 for details. These findings suggest that our shortage index contains additional information beyond that captured by traditional macroeconomic indicators.

Robustness to Individual Newspapers

One concern with an index built by pooling several newspapers is that it may partly reflect the editorial idiosyncrasies of a single paper, or disproportionately capture regional events with little aggregate significance. We address this concern by constructing the shortage index separately for each of the six newspapers in our sample and comparing these single-newspaper indexes to one another and to the pooled headline index. If the headline index were driven by one paper's idiosyncratic coverage, the single-newspaper indexes would diverge sharply from each other and from the aggregate. Instead, we find that they closely track both the headline index and each other: the smallest of the 15 pairwise correlations across the six newspaper-specific indexes is 0.81, and the largest is 0.96. Figure A.6 plots each newspaper's own index (blue) against the headline index (red); the two lines move together in every panel,

Figure A.4: Relationship between Shortages and Inflation and Activity 8-Quarter Ahead



Note: Standardized coefficients from quarterly predictive regressions of inflation and activity on the shortage index. The dependent variable is the log difference between $t+8$ and t of the variable in each row (the price deflator for inflation, real quantities for growth). Controls are the quarterly changes of the dependent variable and of the associated activity or price measure, contemporaneously and with two lags. The full sample runs from 1950:Q1 to 2024:Q4, with subsamples 1950:Q1–1999:Q4 and 2000:Q1–2024:Q4. Shaded bars are heteroskedasticity-consistent 90 percent confidence intervals.

including during the major episodes that drive the headline series. The aggregate index therefore reflects a pattern of shortage coverage shared across newspapers.

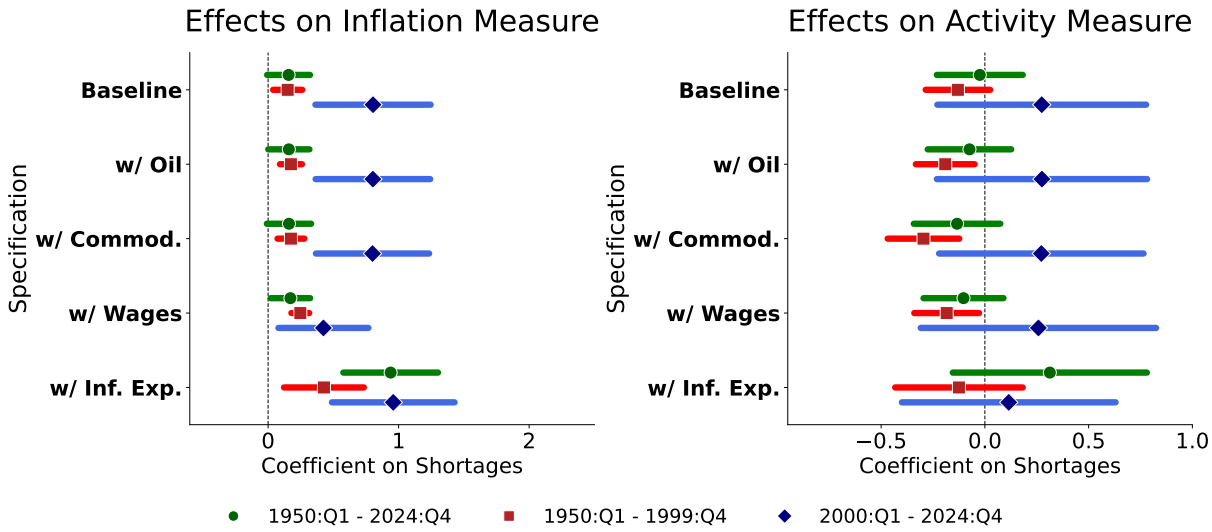
D Additional VAR Results

Priors on Lag Coefficients and Shock Volatilities

We set the priors for the lagged structural coefficients to ensure the model dynamics are empirically grounded while allowing the data to provide substantial updating. Specifically, we estimate an auxiliary VAR(2) on the transformed variables $\mathbf{A}_0(\bar{\theta})\mathbf{X}_t$, where the contemporaneous matrix $\mathbf{A}_0$ is evaluated at the prior means of the structural parameters $\{\delta, \kappa, \alpha, \theta_S, \theta_D\}$. We set the prior means for the diagonal elements of the structural lag matrices $\mathbf{B}_j$ to the OLS point estimates derived from this regression. This procedure centers the prior in a region of the parameter space consistent with the observed persistence of the data, while avoiding draws that imply unstable dynamics. The use of OLS-based dispersion ensures that the prior remains diffuse, so that the likelihood of the data is the primary driver of posterior inference.

To formalize these priors, let $\rho_{j,km}$ denote the structural coefficient on the j-th lag of variable m in the equation for variable k, where the subscripts $k, m \in \{\pi, Y, H\}$ correspond to the variables π_t, y_t, and h_t. These scalars constitute the elements of the lag vectors $\mathbf{b}^S$, $\mathbf{b}^D$, and $\mathbf{b}^H$. For example, the lagged terms in the aggregate supply equation can be written as $[\mathbf{b}^S]'\mathbf{z}_{t-1} = \sum_{j=1}^{p}(\rho_{j,\pi\pi}\pi_{t-j} + \rho_{j,\pi Y}y_{t-j} + \rho_{j,\pi H}h_{t-j})$. The off-diagonal (cross-lag) coefficients are assigned prior means of zero with a standard deviation of 0.05, shrinking the model toward

Figure A.5: Effects of Shortages on Inflation and Economic Activity 4-Quarter Ahead, with and without controls



Note: Standardized coefficients from quarterly predictive regressions of inflation and activity on the shortage index, 4 quarters ahead. Each row shows how the coefficients change as the listed controls are added. The full sample runs from 1950:Q1 to 2024:Q4, with subsamples 1950:Q1–1999:Q4 and 2000:Q1–2024:Q4; regressions with inflation expectations (Federal Reserve Bank of Cleveland, via FRED) start in 1982:Q1 in the earlier subsample. Shaded bars are heteroskedasticity-consistent 90 percent confidence intervals.

own-lag dynamics, in line with standard Minnesota-type priors. The sole exception is $\rho_{1,\pi H}$, for which we set the prior mean at 0.10.

We set the prior means for the standard deviations of the structural shocks, σ_i, to the estimated standard deviations of the residuals from the auxiliary VAR, and assume an inverse-gamma distribution, $\sigma_i \sim IG\left(\hat{\sigma}_i^{OLS}, \frac{1}{2}\hat{\sigma}_i^{OLS}\right)$, where $\hat{\sigma}_i^{OLS}$ is the OLS estimate. This specification provides a centered but relatively diffuse prior for the volatility of the structural innovations. It also accommodates heavy tails by construction: marginalizing the Gaussian structural innovations over an inverse-gamma prior on their variances yields Student-t marginal distributions for the innovations themselves.

VAR Results: Additional Detail

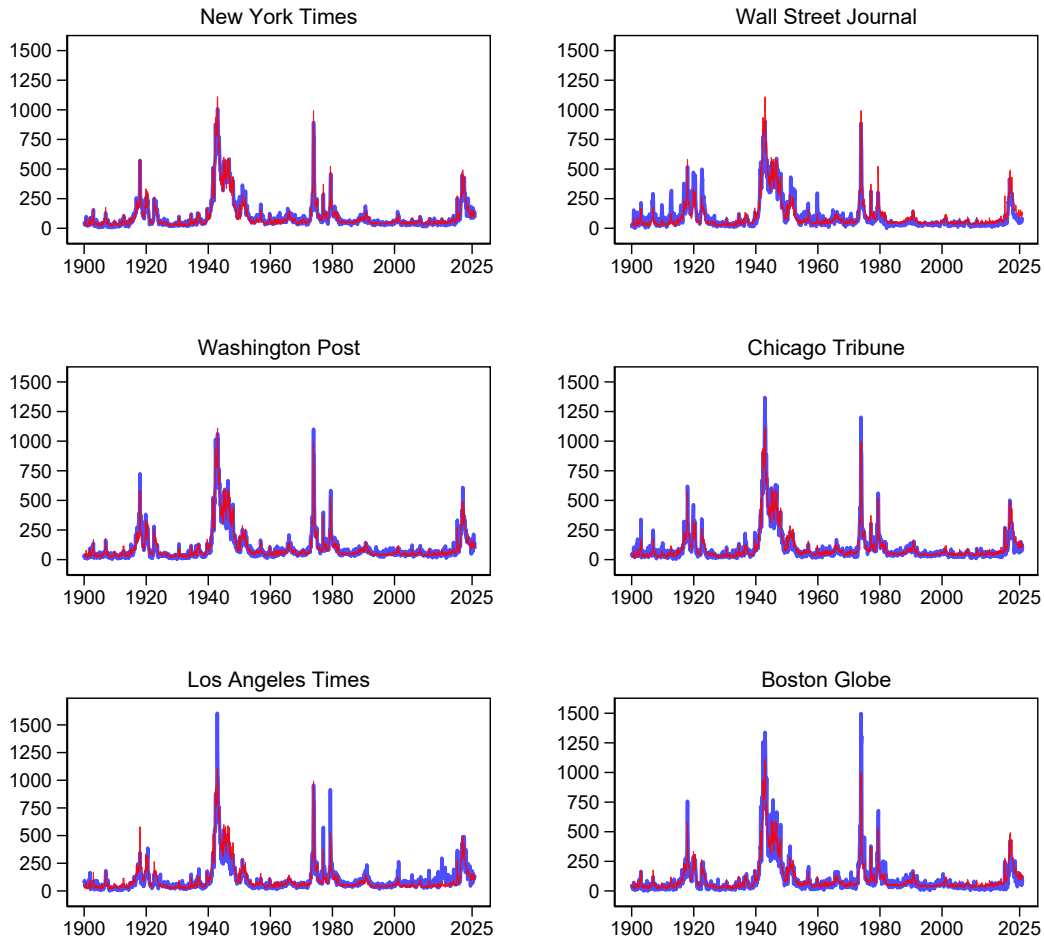
Figure A.7 shows the prior and posterior densities for the model's key structural parameters. Table A.2 provides the same detail, along with posterior mean, standard deviation, and 80 percent credible intervals, for all model parameters.

Robustness to Monetary Policy

This appendix reports details of the VAR model in which we add the policy rate.

We augment the baseline model with the policy rate r_t, measured as the 3-month Treasury bill rate, except during the zero-lower-bound period 2009:Q1–2021:Q4, when we splice in the

Figure A.6: The Shortage Index by Newspaper



Note: Each panel plots the shortage index built only from the named newspaper (blue) against the headline index pooling all six newspapers (red), on a common scale (1900–2019=100). Pairwise correlations among the six newspaper indexes range from 0.81 to 0.96.

Wu and Xia (2016) shadow federal funds rate in its place. The series is expressed in levels, as a deviation from its sample mean, consistent with the other variables in the system:

$$\mathbf{X}_t = (\pi_t, y_t, h_t, r_t)'. \tag{8}$$

The model adds a monetary policy shock u_t^M to the shock vector, $\mathbf{u}_t = (u_t^S, u_t^D, u_t^H, u_t^M)'$. The lag vector $\mathbf{z}_{t-1}$ is extended to include lags of r_t. The contemporaneous structural relationships

Table A.2: VAR Model: Estimated Parameters

		Prior		Posterior			
Parameter	Type	**Mean**	St. Dev	**Mean**	St. Dev	10%	90%
α	Beta	**0.25**	0.12	**0.18**	0.07	0.09	0.27
δ	Inv. Gamma	**0.20**	1.00	**0.13**	0.06	0.07	0.21
κ	Inv. Gamma	**0.20**	1.00	**0.12**	0.03	0.08	0.16
θ_D	Inv. Gamma	**0.20**	1.00	**0.06**	0.01	0.05	0.08
θ_S	Inv. Gamma	**0.20**	1.00	**0.19**	0.05	0.13	0.25
$\rho_{1,HH}$	Normal	**0.84**	0.06	**0.83**	0.04	0.78	0.88
$\rho_{1,H\pi}$	Normal	**0.00**	0.05	**0.05**	0.03	0.01	0.09
$\rho_{1,HY}$	Normal	**0.00**	0.05	**0.04**	0.02	0.01	0.06
$\rho_{1,\pi H}$	Normal	**0.10**	0.05	**0.15**	0.03	0.11	0.19
$\rho_{1,\pi\pi}$	Normal	**0.59**	0.06	**0.60**	0.04	0.54	0.65
$\rho_{1,\pi Y}$	Normal	**0.00**	0.05	**-0.02**	0.03	-0.06	0.01
$\rho_{1,YH}$	Normal	**0.00**	0.05	**0.05**	0.04	-0.00	0.11
$\rho_{1,Y\pi}$	Normal	**0.00**	0.05	**-0.01**	0.04	-0.07	0.04
$\rho_{1,YY}$	Normal	**0.10**	0.06	**0.07**	0.04	0.02	0.12
$\rho_{2,HH}$	Normal	**0.07**	0.06	**0.03**	0.04	-0.02	0.07
$\rho_{2,\pi\pi}$	Normal	**-0.05**	0.06	**-0.03**	0.04	-0.08	0.01
$\rho_{2,YY}$	Normal	**0.15**	0.06	**0.12**	0.04	0.07	0.16
σ_{ε_d}	Inv. Gamma	**1.11**	0.55	**1.08**	0.04	1.02	1.13
σ_{ε_h}	Inv. Gamma	**0.45**	0.23	**0.45**	0.02	0.42	0.47
σ_{ε_s}	Inv. Gamma	**0.63**	0.31	**0.58**	0.02	0.55	0.61

Note: The table shows estimated parameters of the VAR model described in Section 5.

are:

$$\pi_t = \kappa\ y_t + [\mathbf{b}^S]'\mathbf{z}_{t-1} + u_t^S, \tag{9}$$

$$y_t + \alpha\ h_t = -\delta\ \pi_t + [\mathbf{b}^D]'\mathbf{z}_{t-1} + u_t^D, \tag{10}$$

$$h_t = \theta_S\ u_t^S + \theta_D\ u_t^D + [\mathbf{b}^H]'\mathbf{z}_{t-1} + u_t^H, \tag{11}$$

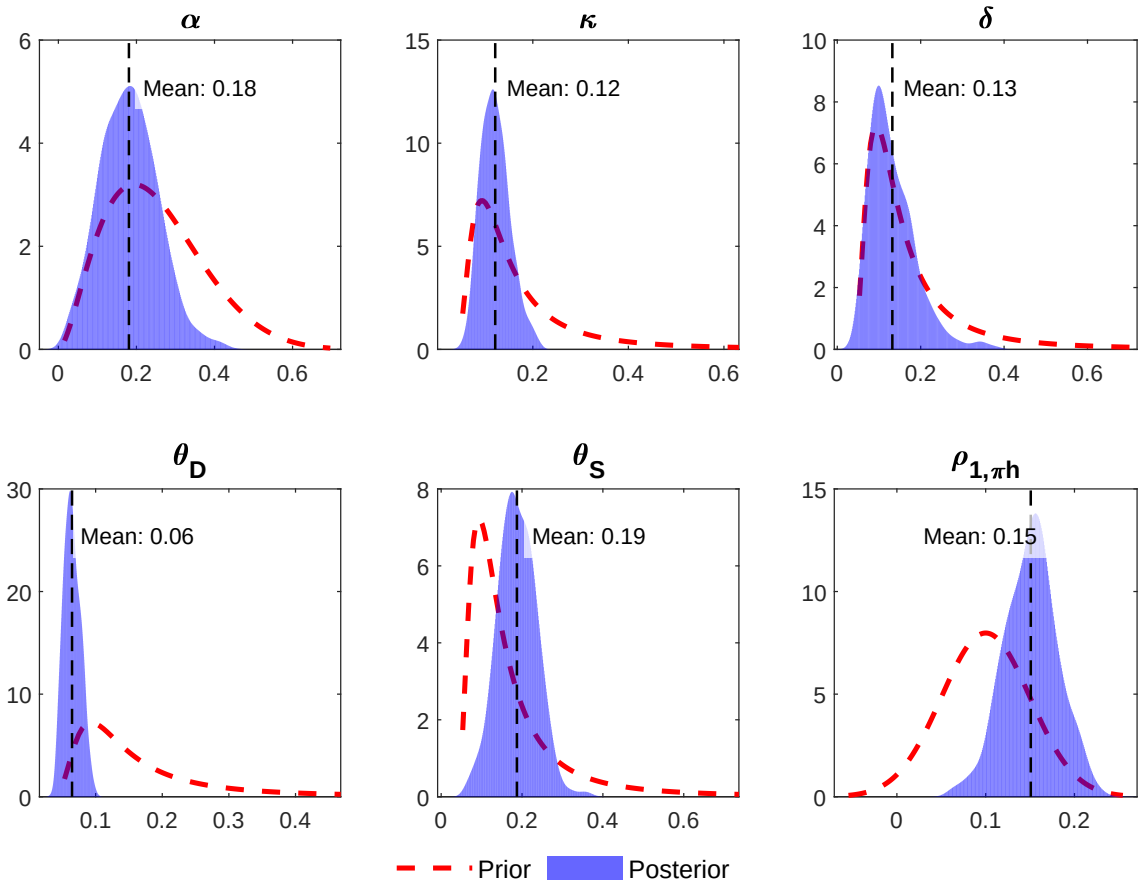
$$r_t = \tau_y\ y_t + \tau_\pi\ \pi_t + [\mathbf{b}^R]'\mathbf{z}_{t-1} + u_t^M. \tag{12}$$

Equations (9)–(11) are identical to the baseline Phillips curve, aggregate demand, and shortage equations of the baseline model, with $\mathbf{z}_{t-1}$ now also carrying lags of r_t. Equation (12) is a Taylor-rule-like policy reaction function: the policy rate responds contemporaneously to output and inflation, with coefficients τ_y and τ_π, plus its own lag dynamics and a monetary policy shock u_t^M. We impose that the policy rate does not respond contemporaneously to shortages directly—only indirectly, through its response to the output and inflation that shortages help determine.[4]

We follow the same Bayesian approach as the baseline model in the main text. The priors for $\kappa, \delta, \theta_S, \theta_D, \alpha$ are unchanged. For the new Taylor-rule coefficients, we assign beta priors with mean 0.10 and standard deviation 0.05 for both τ_y and τ_π, reflecting a modest, disciplined policy response to output and inflation. The new lag coefficients connecting the policy rate to

[4] We omit the $\tau_h\ h_t$ term from equation (12) and fix the corresponding coefficient at zero.

Figure A.7: Prior and Posterior Densities of Structural VAR Parameters



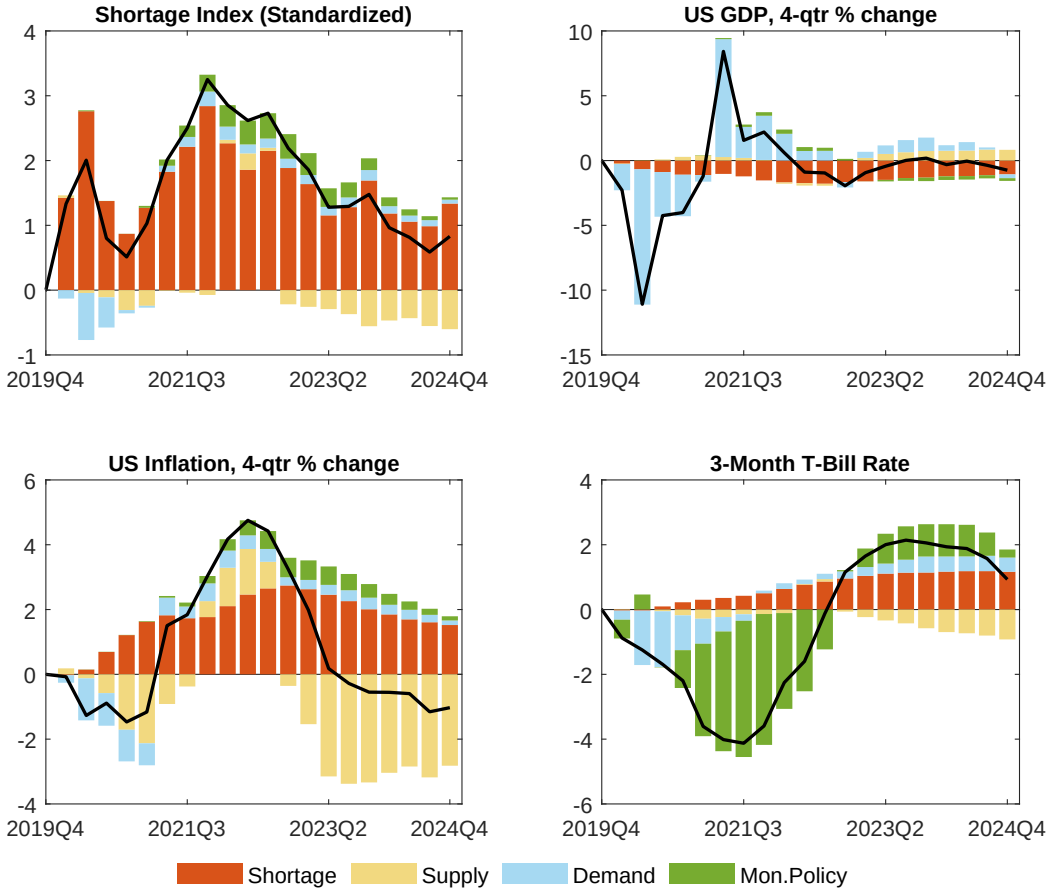
Note: Prior and posterior densities of selected parameters of the structural VAR model of Section 5. Table A.2 reports prior and posterior moments for all parameters.

the rest of the system—the response of shortages, inflation, and output to lagged interest rates, and vice versa—follow the same auxiliary-VAR procedure used for the baseline lag coefficients: we estimate an auxiliary VAR(2) on the transformed variables $\mathbf{A}_0(\bar{\theta})\mathbf{X}_t$ evaluated at the prior means of the structural parameters, and set prior means for the diagonal (own-lag) elements of $\mathbf{B}_j$ to the resulting OLS point estimates. The coefficients on last quarter's interest rate in the shortage, output, and inflation equations are assigned beta priors.[5] We estimate the model using the same algorithm as the baseline.

The posterior estimates of the parameters governing the shortage-supply-demand block are close to their baseline counterparts: the posterior means of κ, δ, θ_S, θ_D, and α are all within 0.03 of their baseline values. The estimated Taylor-rule coefficients are economically sensible: the policy rate responds positively to both output ($\hat{\tau}_y \approx 0.13$) and inflation ($\hat{\tau}_\pi \approx 0.10$), each close to its prior mean of 0.10. Figure A.8 reports the historical decomposition of shortages,

[5] We restrict last quarter's interest rate to only ever reduce—never raise—shortages, output, and inflation, matching the economically intuitive effect of a rate increase. The estimated posterior means for these three coefficients lie well inside the bounds of the beta distribution, indicating that this restriction is not binding in practice.

Figure A.8: Historical Decomposition with Monetary Policy: 2020–2024



Note: Contributions of the four structural shocks—supply, demand, shortage, and monetary policy—to the model's endogenous variables since 2020, in the specification of Appendix D that adds the policy rate. Variables are in deviation from their 2019:Q4 values, with shocks before 2019:Q4 set to zero.

output, and inflation into the four structural shocks over the post-pandemic period. The contributions of supply, demand, and shortage shocks are indistinguishable from the baseline decomposition; the monetary policy shock contributes little to the evolution of shortages, output, or inflation over this period. We conclude that controlling for monetary policy does not alter our reading of the role of shortages in the post-pandemic inflation episode.